CardioAccess AI - A Multimodal Domain-Adaptive AI System for ECG and Echocardiogram-Based Cardiac Screening

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Software Design Specification (SDS)

BY

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Revision history is maintained below to track changes made to this document by any team member.

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1.1	Sahil Kumar	November 2026	Revised the software design to align with the updated proposal and SRS v2.4. Refined the on-demand Referral Summary Generation design, including architecture, API, domain and class models, database schema, security policies, sequence diagram, interface design, and test cases.

Document Sign-Off

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1. Introduction

1.1. Purpose of Document

CardioAccess AI Software Design Specification (SDS) The technical design and implementation architecture of CardioAccess AI. CardioAccess AI is a multimodal, domain-adaptive, and uncertainty-aware web application designed to aid in initial cardiac screening by applying electrocardiogram (ECG) recordings and echocardiography (Echo) videos in resource limited primary-care settings. Functionality of the system is specified in the Software Requirements Specification (SRS v2.4), and in this document we delineate the system architecture, domain and class models, database design, execution time interactions, user-interface structure, security mechanisms and test cases through which these expectations will be met. In this design we describe two parts of this project that work together:

- **The deployed web application -** A frontend in React (Vite with TailwindCSS), a separate FastAPI backend powered by ONNX Runtime for model inference, and Supabase Cloud for authentication, PostgreSQL persistence with Row-Level Security (RLS), and file storage. The application also supports clinician-initiated, template-based Referral Summary Generation for cases with at least one successfully completed screening analysis.
- **The offline model-development pipeline -** Dataset preparation, model training, domain adaptation, active learning, uncertainty and calibration analysis, multimodal fusion, model compression, ONNX export. The resulting versioned model artifacts are used by the deployed application.

This document is intended for this project supervisor and co-supervisor, the development team, the university review committee, and future developers who may want to keep, extend, or repurpose the system. It preserves design traceability between the use cases UC-01 to UC-09 described in SRS v2.4 and functional requirements FR-01 to FR-50. Open issues and “to be confirmed” items are labeled as design decisions relying on not-yet-resolved requirements.

Note: CardioAccess AI is an early-stage research prototype and feasibility study. This is NOT a clinically validated or regulatory approved medical device. Its outputs (screening and generated Referral Summaries) are there to aid professional review and do not indicate a diagnosis or treatment plan. A Referral Summary will be generated only when an authenticated clinician explicitly requests it through the application.

1.2. Definitions, acronyms, and abbreviations

The following terms, acronyms and abbreviations are used throughout this document.

Table 1: Definitions, Acronyms and Abbreviations

Term / Acronym	Definition
AI / DL	Artificial Intelligence / Deep Learning (multi-layer neural networks).
API / REST	Application Programming Interface / Representational State Transfer — the HTTP/JSON interface between the React frontend and the FastAPI backend.
ARRDB	VitalDB Arrhythmia Database; candidate external ECG validation dataset.

Term / Acronym	Definition
AUC / PR-AUC	Area under the ROC curve / area under the precision–recall curve.
CAMUS	Cardiac Acquisitions for Multi-structure Ultrasound Segmentation; external Echo validation dataset (NIfTI sequences).
CAM / Grad-CAM	Class Activation Mapping / Gradient-weighted Class Activation Mapping; saliency methods that show which input regions contributed to a prediction.
Case	A pseudonymous screening record that groups one ECG recording and/or one Echo video submitted by a user, together with all results.
CORAL	Correlation Alignment; unsupervised domain adaptation that aligns second-order feature statistics between source and target domains.
CPEIC	Chaudhry Pervaiz Elahi Institute of Cardiology, Multan — source lineage of the locally sourced ECG-image dataset (to be verified, OI-7).
ECE	Expected Calibration Error — gap between predicted confidence and observed accuracy.
ECG / Echo	Electrocardiogram / Echocardiography (cardiac ultrasound).
ED / ES	End-Diastole / End-Systole — the cardiac phases with maximum and minimum left-ventricular volume.
EDV / ESV / EF	End-Diastolic Volume / End-Systolic Volume / Ejection Fraction, EF = (EDV − ESV) / EDV × 100.
ERD	Entity Relationship Diagram.
Fusion (gated)	A learned network that combines ECG and Echo representations with per-sample gate weights expressing reliance on each modality.
JWT	JSON Web Token — the signed access token issued by Supabase Authentication and verified by FastAPI.
KD	Knowledge Distillation — training a compact student model to mimic a larger teacher.
LV	Left Ventricle.
MC Dropout	Monte Carlo Dropout — repeated stochastic forward passes with dropout active, used to estimate predictive uncertainty.
MIMIC-IV-ECG / -Echo	Credentialed PhysioNet datasets used for patient-paired multimodal fusion experiments.
ONNX / ONNX Runtime	Open Neural Network Exchange model format and its optimized inference engine.
PCP	Primary Care Provider.
PTB-XL	Large public 12-lead ECG dataset; primary ECG training dataset.
RBAC	Role-Based Access Control (Clinician / Nurse / Admin).
RLS	Row-Level Security — PostgreSQL policies that restrict which rows a database user may read or write.
SDS / SRS	Software Design Specification (this document) / Software Requirements Specification.
SE-ResNet1D	One-dimensional residual CNN with Squeeze-and-Excitation channel attention; proposed ECG classifier.
Service role	Privileged Supabase key used only by the backend; it bypasses RLS and therefore requires explicit authorization checks in code.
Signed URL	Time-limited URL that grants temporary read access to a private Storage object.

Term / Acronym	Definition
SUS	System Usability Scale — 10-item usability questionnaire used in the pilot.
TBC	To Be Confirmed — a value that depends on an SRS open issue.
Triage category	Screening-support output: Normal, Needs Review or Urgent Referral. It is not a diagnosis.
TTA	Test-Time Adaptation.
τ (tau)	Uncertainty threshold above which a result is flagged for review (τ_ECG, τ_EF, τ_F); set during validation (OI-4).
U-Net (lightweight)	Encoder–decoder segmentation network with a MobileNetV3 or EfficientNet-B0 encoder, used for LV segmentation.
Referral Summary	Structured record containing a generated Referral Summary and its associated metadata.
Referral Summary Builder	Backend component that fills the predefined summary template using available case and analysis data.
Referral Summary Repository	Data-access component responsible for saving and retrieving Referral Summary records from Supabase PostgreSQL.
Referral Summary Service	Backend service that validates the request, authorizes case access, retrieves the available analysis results, generates the summary, and records the operation.
Template Version	Identifier of the predefined summary-template version used to generate a particular summary.

2. Overall System Description

2.1. Project Background

Cardiovascular disease (CVD) is a leading cause of morbidity worldwide but is mostly associated with low- and middle-income regions where access to a specialist cardiac assessment is limited, as illustrated by examples from the Khyber Pakhtunkhwa region: 1 out of every 200 cardiologists in the country has a credential of certified cardiology, serving a population size over 240 million. CardioAccess AI project is developing compact, domain-adaptive AI methods to triage and address cardiac diseases in resource-poor settings.

Deep learning has displayed efficacy with regards to electrocardiograms (ECG) for detecting cardiac rhythm and ischemic patterns and echocardiography (Echo) for assessing cardiac architecture and performance, including left ventricular ejection fraction (EF). But a systematic review of more than 60 studies from 2019 to 2026 noted that the majority of cardiac AI models have been developed and validated using data from a single institution, patient population, and equipment type. Their performance can degrade when used in a different hospital, country, or population. Of the available studies, the two closest comparator studies which are considered to pinpoint the following research gaps are Gupta et al. (2026) for ECG and Nazari et al. (2025) for Echo. The project therefore addresses seven gaps identified in the literature:

1. Cross-modal (ECG plus Echo) robustness to domain shift is not covered by the comparator studies.
2. Computational complexity and hardware implementation on low-power hardware are not reported in the comparator studies.
3. Confidence calibration under domain shift is underexplored in the comparator studies.
4. No information is provided re: label-efficient adaptation under clinic realistic data sizes in the comparator studies.
5. Validation is conducted in entirety using single-institution datasets.
6. No uncertainty-guided active learning for ECG and Echo adaptation was performed in the comparator studies.
7. No human-factors or usability evaluation was shown in the comparator studies.

It is important to understand the preliminary nature of this novel type of system, where the field still has no agreed definition or regulation and this is just a proof-of-concept. The project is for a research prototype and feasibility study It will result in a working, explainable web application that, after a formative usability pilot with volunteer primary-care providers, is intended to be tested and assessed for its usability and perceived trustworthiness without replacing professionals clinical judgment or providing a clinical diagnosis The system is designed to display screening results with confirmatory confidence component shown with aligned outputs of existing evidence-based sequences, to support initial assessment and referral decisions.

The deployed application supports the generation of structured referral summaries on-demand in addition to its cardiac screening and result-visualization functions. Once a relevant cardiac screening analysis has successfully completed, an authenticated, active and authorized Clinician can request a structured referral summary for a chosen case. It compiles the available case information, which is pseudonymous, with the completed analysis outputs using a pre-defined deterministic template. The summary is then saved in Supabase PostgreSQL, along

with its relevant case and generation metadata, and the application receives it back for confirmation by the Clinician. This feature does not create a different diagnosis or new analysis results, confidence or uncertainty outputs, or change the triage outcome.

2.2. Project Objectives

The objectives below are taken from the SRS (Section 2.4) and the revised proposal. The project will:

1. Systematic review of AI applications in cardiology (2019–2026) examining ECG-based arrhythmia classification and Echo-based ejection fraction (EF) estimation to define its prevalence and consistency in cross-institutional external validation.
2. Design and train an ECG-based cardiac screening classification model using a four-class taxonomy (Normal / Arrhythmia / Ischemic (MI-related) / Other Abnormal), and evaluate its cross-institutional generalization gap through external validation on datasets released by different sources that can be made compatible.
3. Develop and train an ECHO-based model for left ventricular (LV) segmentation and EF estimation, including reduced-EF screening, and determine its cross-institutional generalization gap using a separately sourced dataset.
4. Evaluate lightweight domain adaptation approaches, such as few shot fine-tuning, CORAL and test-time adaptation that could restore cross-institutional performance in models 1 and 2, along with the label-data requirements of each.
5. Implement uncertainty guided active learning to adaptively sample informative local samples to adapt and test against random selection.
6. Test model confidence calibration to assess quality of model under domain shift cross-institution and investigate its level of calibration etc1 before and after adaptation.
7. We will then learn a gated multimodal fusion model using true patient-paired records, and compare its performance and robustness to domain shift to rule-based fusion and individual modality models.
8. Perform model compression and optimize the ECG and Echo models, along with the fusion inference component, for water consumption by implementing efficient compression and deployment solutions and benchmark accuracy, latency, memory usage and computational feasibility for hardware typical of resource-limited primary-care settings.
9. Appraise the usability and perceived trustworthiness of a working web prototype that integrates the evaluated models and the adaptation, uncertainty, calibration, fusion and explainability components through a structured pilot using non-specialist's healthcare workers or primary-care providers. The expected outcomes are (a) measured findings on cross-institutional generalization, domain-adaptation recovery, confidence calibration, multimodal fusion and computational efficiency, and (b) a working, explainable, browser-accessible web application. Table 2 shows how each objective is addressed in this design.

Table 2: Objective-to-Design Traceability

#	Objective (short)	Design elements that realize it	SDS section
1	Literature review	Literature review and comparator-study discussion inform the project background and model-development direction.	2.1

#	Objective (short)	Design elements that realize it	SDS section
2	ECG classifier + gap	Offline pipeline; ECGPipeline, ECGPreprocessor, ECGClassifier; ecg_results table.	3, 5, 6
3	Echo segmentation + EF	EchoPipeline, EchoSegmenter, VolumeEstimator; echo_results table.	3, 5, 6
4	Domain adaptation	FusionModel; inference-orchestrator pairing check; Fusion results table stores fused results and gate weights.	3.2, 5, 8
5	Active learning	Uncertainty Estimator scores drive offline sample selection.	3.1, 5
6	Calibration	Calibration reports (ECE, Brier, reliability) in offline evaluation; thresholds τ feed Triage Engine.	3.2, 10
7	Gated fusion	FusionModel, pair_eligible() rule, fusion_results with gate weights.	3.2, 5, 8
8	Compression + benchmark	ONNX model artifacts; export-parity and quantization checks (TC-43); latency and memory benchmarking (TC-45).	3.5, 10
9	Prototype + usability	React frontend, FastAPI backend and Supabase; structured usability pilot with volunteer PCPs, recording task times, SUS scores and trust feedback (TC-47)	3, 9, 10
10	Referral Summary Generation	Clinician-triggered React action, ReferralSummaryService, ReferralSummaryBuilder, referral_summaries table, authorization, audit logging, and summary review interface.	3, 4, 5, 6, 7, 8.8, 9.5, 10

3. System Architecture

CardioAccess AI is built on a client-server, multi-tier architecture split into run-time application and offline research pipeline (Figure 1). The browser never runs any AI models nor holds privileged credentials; the FastAPI backend is the only component that is able to instantiate patients files, run ONNX models and save results; Supabase Cloud provides managed identity, relational storage and object storage. The offline pipeline creates versioned model artifacts that are loaded into the model-serving layer.

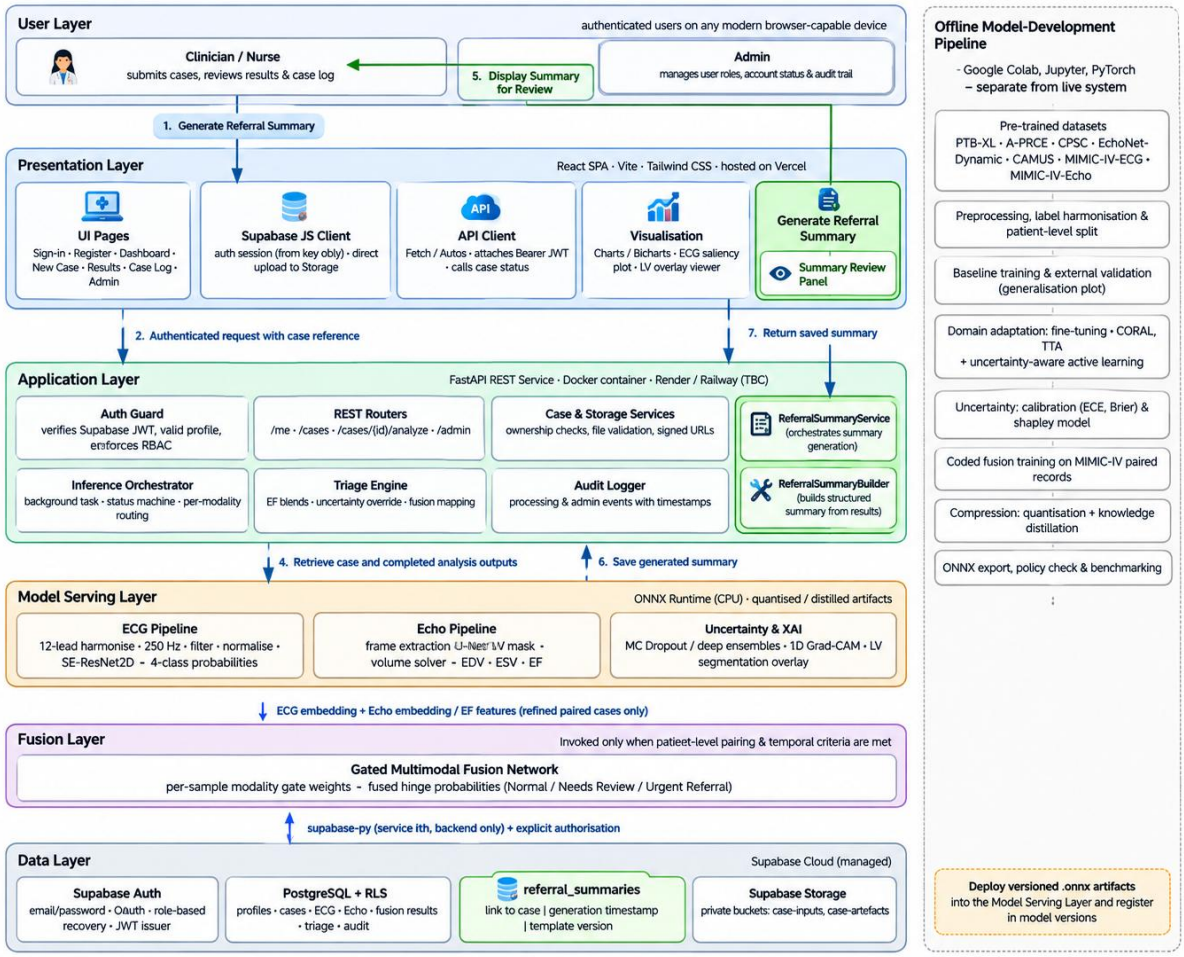


Figure 1: System Architecture of CardioAccess AI

3.1. Architectural Layers and Components

The Case Result and Explanation screen contains a Referenced Summary (Generate Referral Summary) action and a read-only Summary Review panel, both of which clinicians view (Reference Summary) and clinicians only can. This will only be revealing to a logged in, active Clinician who is authorized to view the selected case, and only after one relevant analysis completed successfully. Select the case reference in React and the user's Supabase JWT that Relay sends to the FastAPI backend. Then the backend authenticates the authentication token, checks the case authorization and then pulls all relevant case and analysis info.

- **User layer:** authenticated users are also the clinicians/nurses (they can submit cases, review test results, and navigate their case logs) and Admins (who handle role

configuration, account status, the audit trail) are required to be users logged in through a modern browser.

- **Presentation layer:** Single-page React application. Authenticates directly with Supabase Auth, uploads input files directly to a private Storage bucket in the users own folder, and calls FastAPI with the user's JWT for every protected operation. Only the public anon key is contained in the frontend bundle. The case-result screen also provides the Generate Referral Summary action. Once the back end returns the summary back to React after it's been saved, it is rendered in a read-only review panel that indicates its generation time, the available modalities, and any unavailable information.
- **Application layer:** A FastAPI service packaged as a Docker container. The AuthGuard dependency, which validates each JWT and loads the user profile. Case and admin endpoints via routers, services provide ownership checks, file validation, background inference orchestration, triage rules, and audit logging. The application layer also contains Referral Summary Service and Referral Summary Builder. ReferralSummaryService validates the authenticated request, checks case authorization, retrieves the available case and analysis records, coordinates summary generation, persists the generated record, and records the operation through AuditLogger. ReferralSummaryBuilder populates a predefined, deterministic template and does not invoke an additional AI model.
- **Model-serving layer:** ONNX Runtime sessions for the SE-ResNet1D ECG classifier and the lightweight U-Net Echo segmenter, with shared uncertainty estimation and explanation generation.
- **Fusion layer:** The gated multimodal fusion network, only when case contains verified patient-paired ECG and Echo inputs (FR-40).
- **Data layer:** The JWT used by the app is issued by Supabase Authentication. Supabase PostgreSQL maintains user profiles, pseudonymous case info, uploaded-file metadata, analysis results, triage decisions, audit events, and generated Referral Summary records. The referral_summaries table holds every summary alongside its case reference, responsible user, relevant analysis references, template version, and generation timestamp. Supabase Storage is responsible for uploaded ECG and echocardiography files, and derived visual artifacts. A PDF summary file is placed in Storage only if PDF export is enabled. Database access is protected by RLS, FastAPI backend is responsible for performing explicit authorization checks before accessing any case data.
- **Offline model-development pipeline:** Runs on Google Colab Pro / Kaggle GPU environments and remains separate from the deployed runtime (FR-39, SRS Section 2.6).

Table 3: Architectural Components and Responsibilities

Layer	Component	Technology	Responsibility	SRS ref.
Presentation	UI pages	React, Vite, Tailwind CSS	Sign-in, registration, dashboard, new case, results, case log, admin screens.	FR-25, UC-01–08

Layer	Component	Technology	Responsibility	SRS ref.
Presentation	Supabase JS client	supabase/supabase-js	Sign-up, sign-in, sign-out, password recovery; direct upload to case-inputs bucket.	FR-30, 31, 34
Presentation	API client	Fetch / Axios	Calls FastAPI with Bearer JWT; polls case status.	FR-33, 39
Presentation	Visualization	Chart.js / Recharts, SVG	Probability bars, ECG saliency, LV overlay, volume curve.	FR-25
Presentation	Summary action and review panel	React	Provides the clinician-triggered Generate Referral Summary action and displays the saved summary for review.	UC-09, FR-41, FR-48
Application	AuthGuard	FastAPI dependency, PyJWT	Verify JWT signature/expiry/audience; load profile; enforce roles.	FR-32, 33
Application	CaseRouter / AdminRouter	FastAPI routers, Pydantic	REST endpoints, request validation, error mapping (Table 5).	FR-23, 29, 32
Application	CaseService	Python, supabase-py	Create cases, authorize ownership, register files, list/filter cases.	FR-27, 29, 35, 37
Application	Storage Service	supabase-py Storage	Validate metadata, download inputs, upload artifacts, create signed URLs.	FR-34, 35, 36
Application	Inference Orchestrator	FastAPI Background Tasks	Status machine, routing per modality, pairing check, persistence.	FR-40, UC-04–06
Application	Triage Engine	Python	EF bands, uncertainty override, fused-category mapping.	FR-08, 16, 24
Application	AuditLogger	Python	Append-only processing and administrative events.	SRS 5.2, 5.3

Layer	Component	Technology	Responsibility	SRS ref.
Application	ReferralSummaryService	Python, FastAPI	Validates the request, checks case authorization, retrieves available case and analysis data, coordinates generation, persists the summary, and records the operation.	UC-09, FR-42, FR-43, FR-46, FR-49
Application	ReferralSummaryBuilder	Python	Populates the predefined deterministic template using available case and analysis outputs. It does not run another model or modify the source results.	UC-09, FR-44, FR-45
Application	ReferralSummaryRepository	Python, Supabase Client	Saves and retrieves Referral Summary records from Supabase PostgreSQL.	UC-09, FR-46, FR-47
Model serving	ECG / Echo pipelines	NumPy, SciPy, wfdb, OpenCV, NiBabel, ONNX Runtime	Preprocessing, inference, uncertainty, explanations.	FR-01, 07, 14, 40
Fusion	FusionModel	ONNX Runtime	Fused probabilities and gate weights.	FR-18, 19, 40
Data	Auth / PostgreSQL / Storage	Supabase Cloud	Identity, relational data with RLS, private object storage.	FR-36–39
Data	referral_summaries	Supabase PostgreSQL	Stores summary content, case reference, generating user, analysis references, template version, and generation timestamp under RLS.	UC-09, FR-46, FR-47

3.2. Run-time Inference Pipeline

Figure 2 illustrates backend processes after sending a case. It analyses in background so the request will return immediately (202 Accepted) and the browser will poll the case status. Each modality is processed in its own lane; it means that if one fails the other does not get discarded and the case will end as Partially Completed. The band-pass 0.5–40 Hz, 10-second ECG window, echo frame size of 112 × 112 are the preprocessing parameters that are tied to the models to be trained by default configuration values.

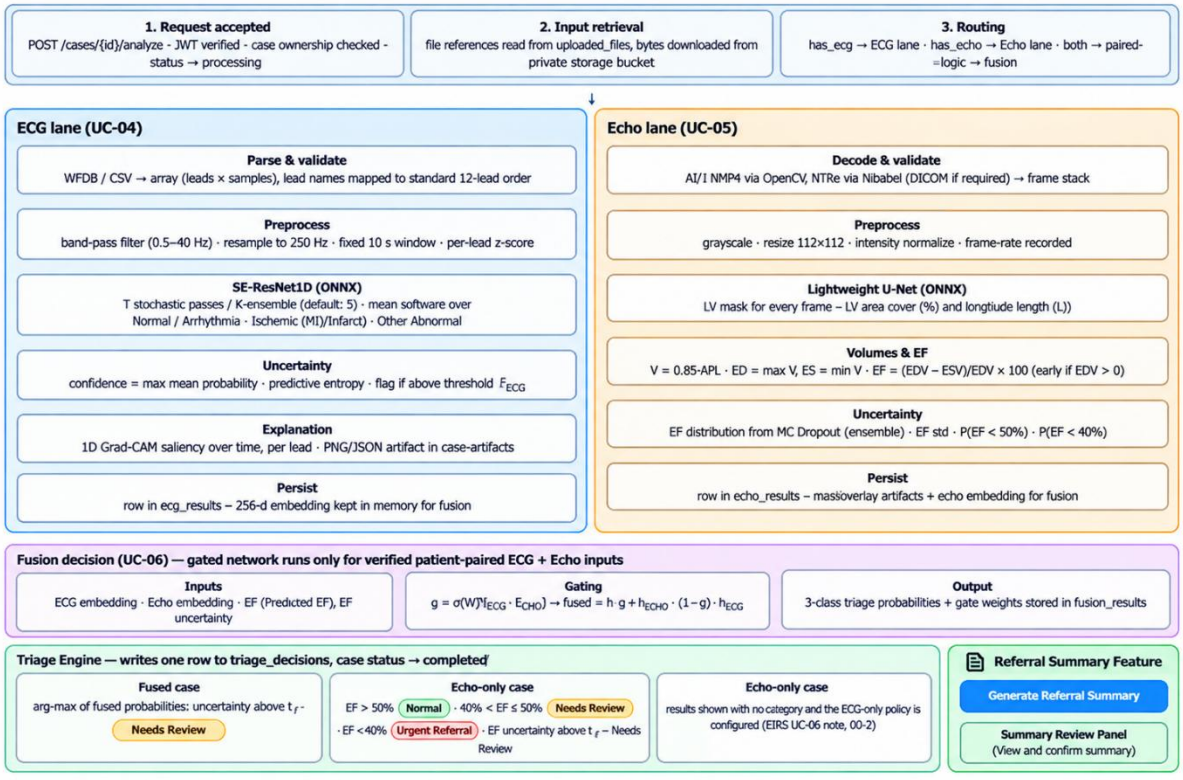


Figure 2: Run-time Inference and Triage Pipeline

Uncertainty in ONNX. ONNX Runtime has no training mode, so predictive uncertainty is captured by one of two selected during validation, as a per model basis: (a) The exported graph keeps dropout stochastic via explicit random masks, while keeping batch-normalization from training (i.e., in inference mode), and run T stochastic passes (MC Dropout); or (b) Export K independently trained ensemble members and average them. I again get a mean prediction, a spread (standard deviation or entropy) and for Echo the probabilities P (EF < 50%) and P (EF ≤ 40%) of the EF values (FR-14 FR-15).

Explanations without autograd. The ECG classifier ends in global-average pooling followed by a linear layer, for which Grad-CAM at the final convolutional layer becomes Class Activation Mapping. Consequently, the last-layer activations are exported from the ONNX graph as an additional output, and Explanation Generator then calculates the one-dimensional signal saliency as a class-weighted sum of these activations - no run-time gradients are required. The Echo explanations are the LV segmentation masks displayed as overlays (SRS Table 18).

Volume estimation. The single-plane area-length method is used by default, LV volume per frame is estimated from the segmented area A and long-axis length L, $V = 0.85 \cdot A^2 / L$; the frame with maximum volume is taken as ED and the minimum as ES. The method and its geometric assumptions remain to be validated (OI-11), so the Volume Estimator is designed, which allows the method of discs to be replaced without changing its interface.

Referral Summary Generation (On demand): Referral Summary Generation is a distinct, clinician-initiated operation performed after the relevant screening analysis has completed. It is available only to an authenticated, active, and authorized Clinician when at least one relevant analysis has completed successfully. Upon a request, ReferralSummaryService acquires the case info and completed ECG, Echo, fusion, confidence, uncertainty, explainability and triage outputs from Supabase PostgreSQL as applicable. Availability of information is organized into the template used by the ReferralSummaryBuilder into a structured referral summary. Unavailable outputs of a modality or an individual analysis are specifically flagged as unavailable instead of being faked or replaced by an interpretation. The generated summary is persisted into the referral_summaries table, with the case reference, sum of data, who persist the data, version of template, relevant analysis references and the generation timestamp. The summary is returned to React for clinicians to audit. As this operation has no impact on the original prediction, confidence or uncertainty estimates, unchanged analysis records, or unchanged triage outcomes, the service will not begin automatic generation, regular processing or automatic escalation.

Table 4: Triage Decision Rules (Triage Engine)

Case type	Condition	Category	Notes
ECG + Echo, verified pair	Highest fused probability	Normal / Needs Review / Urgent Referral	source = fusion; gate weights stored (FR-19). Fused category mapping and decision thresholds remain subject to OI-2.
ECG + Echo, verified pair	Fused uncertainty > τ_F	Needs Review	Overrides the arg-max category (FR-16).
Echo only (or pair not verified)	EF > 50%	Normal	source = echo (FR-08, FR-24).
Echo only	40% < EF ≤ 50%	Needs Review	
Echo only	EF ≤ 40%	Urgent Referral	
Echo only	EF std > τ_{EF} or low-confidence flag	Needs Review	Regardless of EF band (SRS 5.2).
Echo only	EF unavailable (segmentation failed / EDV ≤ 0)	No category	Measurements shown as unavailable, never invented.
ECG only	No ECG-only policy configured	No category	ECG probabilities and flags are still displayed (UC-06 note, OI-2).

3.3. REST API Design

All endpoints are versioned under `/api/v1`, exchange JSON, and require `Authorization: Bearer <JWT>` except the health check. Requests for cases the caller does not own return **404** rather than 403 so that the existence of other users' cases is not revealed.

Table 5: REST API Endpoints

Method & path	Role	Purpose	Success
GET /health	Public	Liveness check and list of loaded model versions.	200
GET /api/v1/me	Any signed-in	Return the caller's profile and role (used after sign-in).	200
POST /api/v1/cases	Clinician / Nurse	Create a draft case; returns case_id, case_code and upload prefix.	201
POST /api/v1/cases/{id}/files	Owner	Register an uploaded file (modality, path, MIME type, size, SHA-256) after validation.	201
POST /api/v1/cases/{id}/analyze	Owner	Queue analysis of all registered files.	202
GET /api/v1/cases	Owner	Paginated case log; filters: status, triage, modality, date range.	200
GET /api/v1/cases/{id}	Owner	Case detail with ECG, Echo, fusion and triage results.	200
GET /api/v1/cases/{id}/status	Owner	Lightweight status for polling.	200
GET /api/v1/cases/{id}/artifacts/{name}/url	Owner	Short-lived (60 s) signed URL for a saliency plot or LV overlay.	200
GET /api/v1/admin/users	Admin	List users with role, status and last sign-in.	200
PATCH /api/v1/admin/users/{uid}	Admin	Change role or activate/deactivate an account (audited).	200
GET /api/v1/admin/audit-events	Admin	Paginated audit trail.	200
POST /api/v1/cases/{id}/referral-summary	Authenticated and authorized Clinician	Generates a structured summary from the available case information and completed analysis outputs, saves it in `referral_summaries`, and returns the saved record.	201

Errors share one JSON shape, `{"error": "<code>", "message": "<text>"}`, with status codes 400 (malformed request), 401 (missing, invalid or expired token), 403 (role not permitted), 404 (case not found or not owned), 413 (file too large), 415 (unsupported type),

422 (validation failure, e.g. no valid file) and 500 (unexpected error; details are logged, not returned).

The referral-summary endpoint requires a valid Supabase JWT and a case reference. FastAPI validates the token and verifies that the requesting clinician is authorized to access the selected case before retrieving any case or analysis information. The endpoint generates a summary only when at least one relevant analysis has completed successfully. It returns the saved summary record after successful persistence. If authentication, authorization, data retrieval, generation, or persistence fails, the backend returns an appropriate error response and does not report the operation as successfully completed.

3.4. Security Architecture

For Referral Summary Generation, the backend validates the Supabase JWT and verifies the requesting clinician's permission to access the selected case before retrieving case information or analysis outputs. The summary is saved in the referral_summaries table and remains linked to its parent case. Row-Level Security restricts access to summary records according to the associated case permissions. The backend also records relevant summary-generation and denied-access events in the application audit log.

- **Authentication:** Supabase Auth email/password with mandatory email verification; sessions are refreshed by the Supabase client (FR-30, FR-31).
- **Token validation:** AuthGuard checks the JWT signature, expiration, and audience when all protected requests happen and rejects inactive profiles (FR-33).
- **Authorization in two places:** RLS policies enforce browser read access. The backend processes through the service role (bypassing RLS) so that each service method calls CaseService.Authorize() before attempting to access a case or file (FR-37).
- **Role assignment:** New profiles are created by a database trigger with the role clinician; the role and is_active columns cannot be updated by authenticated users, so Admin privileges cannot be self-assigned (FR-32).
- **Secrets:** Service role keys and database credentials are stored solely as backend environment variables; the frontend bundle holds only the public anon key (FR-38).
- **Files:** Inputs are stored in the private case-inputs bucket under {user_id}/{case_id}/.; generated artifacts are placed into the private case-artifacts bucket and can only be retrieved via 60-second signed URLs (FR-34–36).
- **Transport and minimization:** HTTPS/TLS everywhere; pseudonymous case codes; free-text notes suppress personal identifiers (SRS 5.3).
- **Auditability:** Submissions, analyses, triage generation, role modifications, and denied access attempts get recorded in the append-only audit_events table.

3.5. Deployment View

Table 6: Deployment Nodes

Node	Hosts	Configuration
End-user device	Browser running the React SPA	Current Chromium-based browser or Firefox; internet required; no local model installation.

Node	Hosts	Configuration
Vercel (frontend)	Static React build	Environment: SUPABASE_URL, SUPABASE_ANON_KEY, API_BASE_URL; HTTPS by default.
Render / Railway (backend) [TBC]	Docker image: FastAPI + ONNX Runtime + .onnx artifacts	Environment: SUPABASE_URL, SUPABASE_SERVICE_ROLE_KEY, JWT secret/JWKS URL, thresholds; resource tier [TBC] (OI-12).
Supabase Cloud	Auth, PostgreSQL (+RLS), Storage	SQL migrations for schema, enums, triggers and policies kept in the repository.
Google Colab Pro / Kaggle	Offline training and evaluation	Credentialed datasets accessed under their data-use agreements; never deployed.

Referral Summary Generation uses the same deployed React frontend, FastAPI backend, and Supabase PostgreSQL services as the rest of the application. It does not require a separate deployment service, background scheduler, or additional machine-learning model. The summary is generated in response to an authenticated clinician request and stored as a structured database record associated with the selected case.

3.6. Key Design Decisions

Table 7: Key Design Decisions and Rationale

Decision	Rationale	Alternative considered
Browser uploads directly to Supabase Storage	Echo videos can reach tens of MB; streaming them through FastAPI would double bandwidth and memory on the backend.	Multipart upload to FastAPI.
Background task + status polling	Inference with T passes may take several seconds; avoids HTTP timeouts with minimal infrastructure.	Celery/Redis queue (heavier than needed for prototype scale).
Service role + explicit authorization	RLS is bypassed by the service role, so ownership is checked in code on every operation (FR-37).	Forwarding the user JWT to PostgreSQL for every query.
Segmentation-based EF	Interpretable pathway (masks → volumes → EF) with visual overlays; direct video-to-EF kept as baseline (FR-09).	Direct regression only.
CAM-style saliency from exported activations	ONNX Runtime has no autograd; equivalent to Grad-CAM for a GAP + linear head.	Running PyTorch on the server.
Fusion only for verified pairs	Fusion was trained on genuinely paired MIMIC-IV records; applying it to unpaired inputs would be unsupported.	Fusing any case that has both files.
404 for non-owned cases	Prevents enumeration of other users' case IDs.	403 Forbidden.
Thresholds as configuration	τ values, upload limits and accepted formats depend on validation results and open issues.	Hard-coded constants.

Decision	Rationale	Alternative considered
Deterministic, predefined summary template; no LLM or additional model	The summary compiles existing stored results in a reproducible manner and does not generate new clinical findings.	LLM-generated narrative; manually authored case-by-case summary.
Clinician-initiated generation only	The clinician reviews the completed results and explicitly requests the summary. This avoids automatic generation after analysis or scheduled background processing.	Automatic generation after analysis; scheduled or background generation.

Table 8: Configurable Design Parameters

Parameter	Default / proposed value	Source
ECG target sampling rate	250 Hz, 12 leads, 10 s window	FR-01
Accepted ECG formats / size	WFDB (.hea + .dat in .zip), CSV with 12 lead columns; ≤ 10 MB [TBC]	FR-23, OI-1
Accepted Echo formats / size	AVI, MP4, NIfTI (.nii, .nii.gz); DICOM if required; ≤ 100 MB [TBC]	FR-23, OI-1
EF bands	Normal > 50%; Needs Review > 40% and ≤ 50%; Urgent ≤ 40%	FR-08, FR-24, OI-3
Uncertainty passes / members	T = 20 MC passes or K = 5 ensemble members	FR-14
Thresholds τ_ECG, τ_EF, τ_F	Set from validation calibration results [TBC]	FR-16, OI-4
ECG-only triage policy	Not configured → no category	UC-06, OI-2
Pairing time window	Defined by fusion data preparation [TBC]	FR-18, OI-10
Signed URL lifetime / poll interval	60 s / 3 s	SRS 5.3
Summary template version	v1.0 (initial template; versioning retained)	
PDF export	Optional; implemented only if included in the final application	

4. Domain Model

The domain model (Figure 3) models the real world concepts in CardioAccess AI and their relationships independent of technology. A User is subspecialized into Clinician/Nurse and Admin. A Case for a Patient is submitted by Clinician/Nurse and each case has 1 or 2 Input Recordings an ECG R & Echo R. Each recording can be analyzed into an Analysis Result, specialised into ECG Analysis (four class probabilities) and Echo Analysis (ED/ES frames, EDV, ESV, EF and reduced-EF probabilities). Each model has a distinct Model Version, and may generate one or more Explanations. The Fusion Result uses both analyses as inputs when they exist as part of a verified pair, otherwise the triage is coming from Echo analysis alone. All important actions are Audit Events.

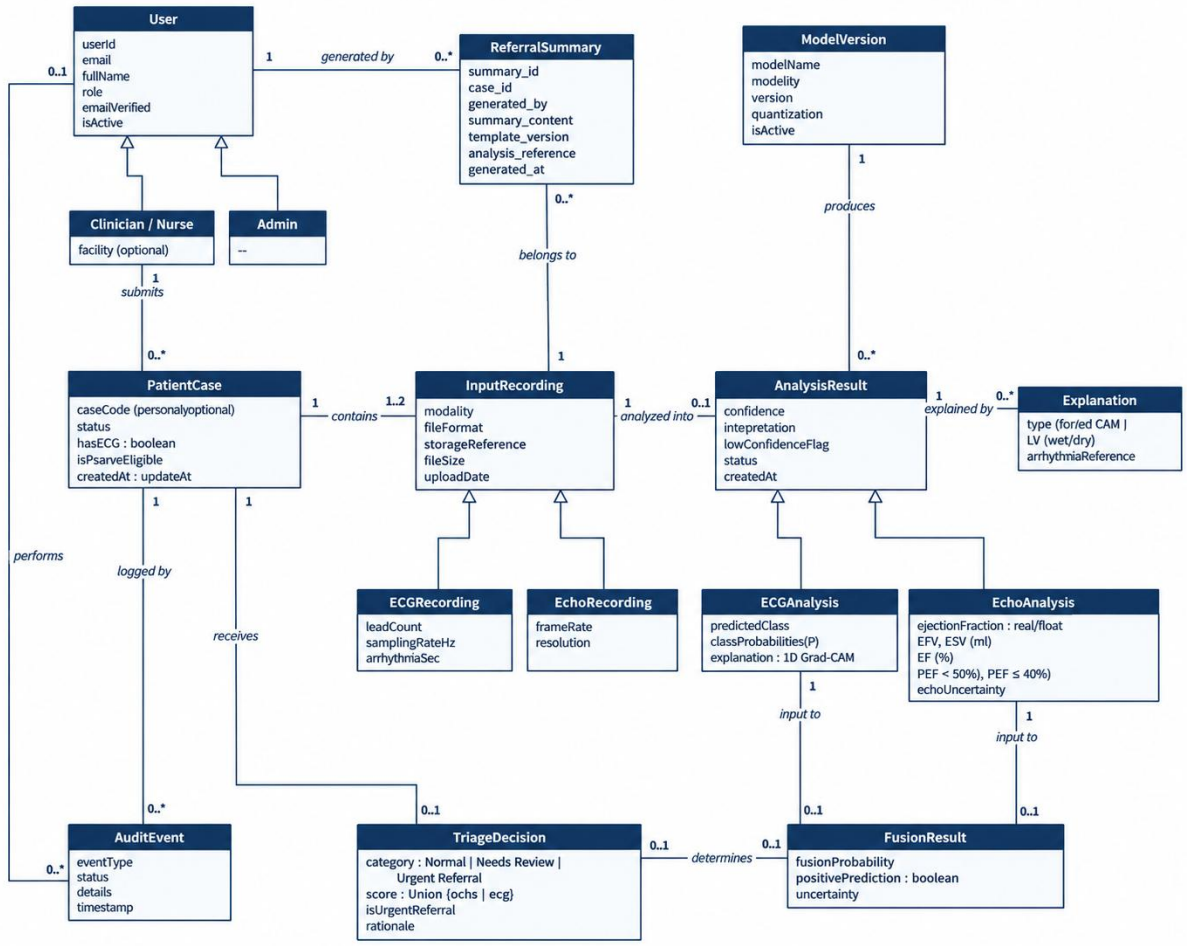


Figure 3: Domain Model of CardioAccess AI

The domain model also includes ReferralSummary, which represents the structured summary generated from available information and completed analysis outputs of a selected case. Each ReferralSummary is associated with one Patient Case and records the authenticated clinician who requested it, the template version used, the relevant analysis references, and the generation timestamp. If a clinician requests to generate more summary records for a case, the case may have many summary records. The summary is a referral-support document and is not a diagnosis in itself or a change to the original triage decision.

Table 9: Domain Concepts

Concept	Description
User	A registered, email-verified person with exactly one role.
Clinician / Nurse	Non-specialist end user who submits and reviews cases (UC-03, UC-07, UC-08).
Admin	Manages accounts, roles and the audit trail; cannot be self-registered (FR-32).
Patient Case	Pseudonymous unit of work with a lifecycle (draft → queued → processing → completed / partially_completed / failed).
Input Recording	Abstract uploaded input; ECGRecording (leads, sampling rate, duration) or EchoRecording (frames, frame rate, resolution).
Analysis Result	Abstract model output with confidence, uncertainty and a low-confidence flag; specialised into ECGAnalysis and EchoAnalysis.
Explanation	Visual artifact — ECG Grad-CAM-style saliency or LV segmentation overlay.
Fusion Result	Fused triage probabilities and modality gate weights for a verified ECG–Echo pair.
Triage Decision	Screening category with its source (fusion / echo / ecg), uncertainty flag and rationale.
Model Version	Identifies the deployed ONNX artifact that produced a result (reproducibility).
AuditEvent	Timestamped record of a processing or administrative action.
Referral Summary	Structured summary generated only when an authenticated and authorized Clinician explicitly requests it. It contains available case information and completed analysis outputs, is linked to one Patient Case, and records the generating user, template version, relevant analysis references, and generation timestamp. It does not constitute a diagnosis.

5. Class Diagram

The backend is built in Python and is organized into to controllers (FastAPI routers), services, repositories, entities and an inference package. Splitting the class diagram into two views helps simplify it for readability. Figure 4 (application level and service level): routers depend on AuthGuard to provide identity and roles that are forwarded to the CaseService and InferenceOrchestrater; services persist entities by using subclasses of the SupabaseRepository; the orchestrator uses the Triage Engine and writes through the AuditLogger. Entity classes are Pydantic models that copy the database tables.

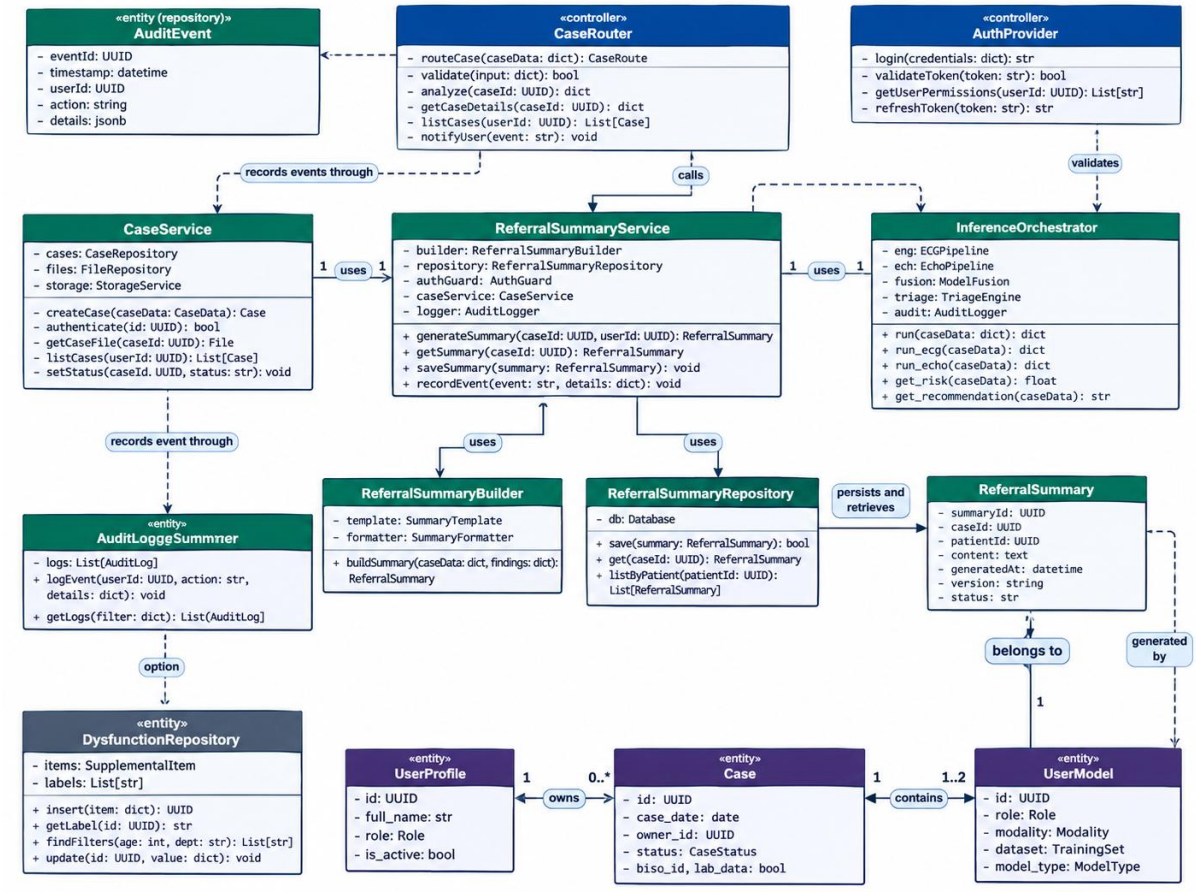


Figure 4: Class Diagram - Application and Service Layer

The inference and model layer by figure is shown in Figure 5. ECGPipeline and EchoPipeline are each composed of a preprocessor as specialization of the abstract Preprocessor and an ONNX model as specialization of the abstract OnnxModel, and they both use the shared Uncertainty Estimator and Explanation Generator. EchoPipeline includes an added VolumeEstimator, which turns per-frame masks into the volume curve, ED/ES frames and EF. FusionModel is a third OnnxModel to be used by the orchestrator. Every model shares the same OnnxModel interface (load, run), and every preprocessor shares its load/transform interface, so compressed or adapted models can simply be registered by adding a new Model Version, no code changes are required.

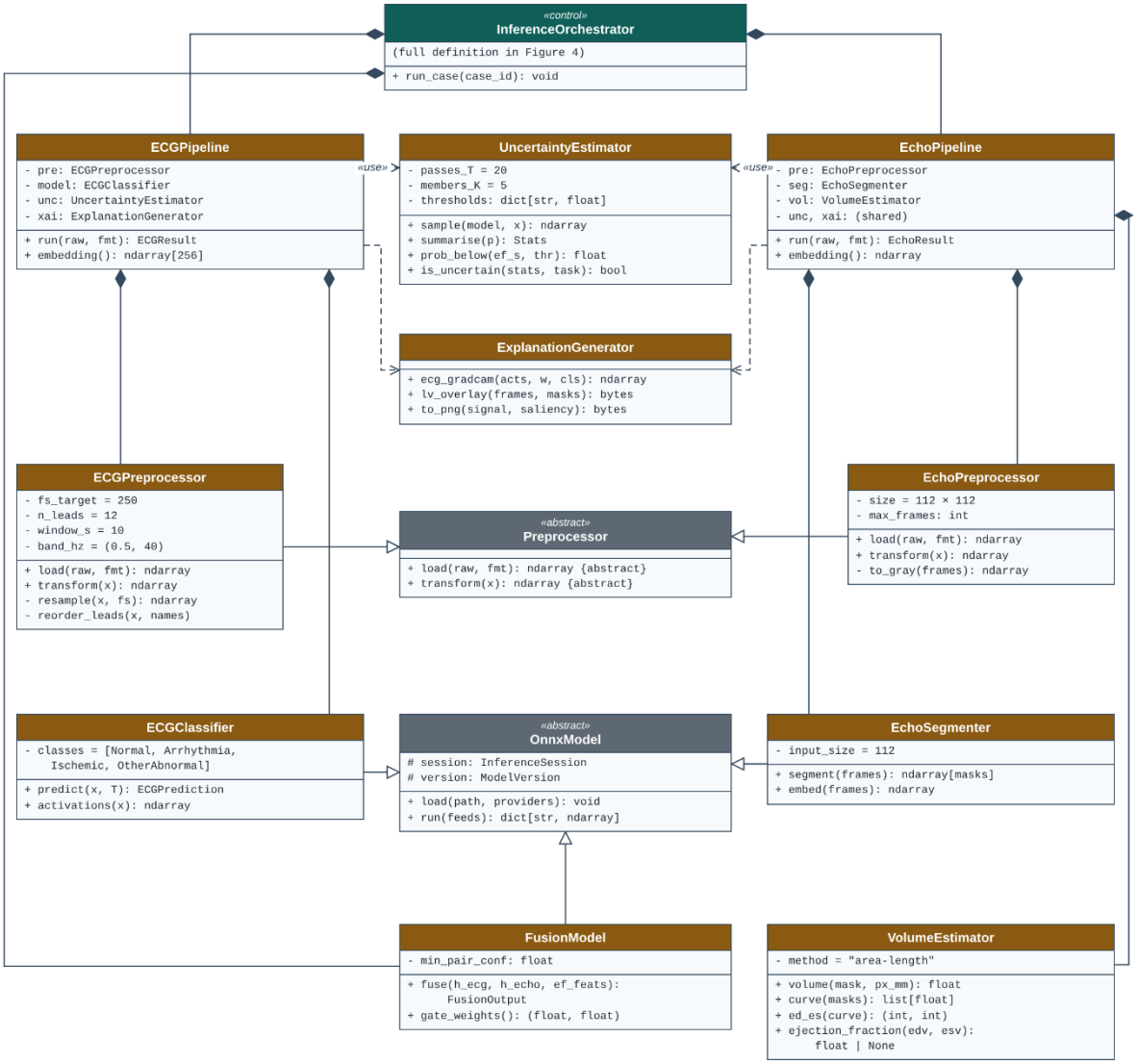


Figure 5: Class Diagram - Inference and Model Layer

Table 10: Class Responsibilities

Class	Responsibility	Traces to
AuthGuard	Verifies Supabase JWTs, returns the current UserProfile and enforces required roles.	FR-32, FR-33
CaseRouter / AdminRouter	Map HTTP requests to service calls; validate request and response schemas.	UC-03, UC-07, UC-08
CaseService	Creates cases with unique codes, authorizes ownership, registers files, lists and filters cases.	FR-27, FR-29, FR-37
Storage Service	Validates type/size, downloads inputs, uploads artifacts, issues signed URLs.	FR-23, FR-34–36
Inference Orchestrator	Runs the case state machine, routes modalities, checks pair eligibility, persists results.	FR-40, UC-04–06
Triage Engine	Applies EF bands, uncertainty override and fused-category mapping (Table 4).	FR-08, FR-16, FR-24
AuditLogger	Writes append-only audit events.	SRS 5.2, 5.3
SupabaseRepository	Base CRUD over a table using the service-role client.	FR-27, FR-36
ECGPreprocessor	Parses WFDB/CSV, reorders leads, filters, resamples to 250 Hz, normalizes.	FR-01, FR-02
Echo Preprocessor	Decodes AVI/MP4/NIfTI, converts to grayscale, resizes and normalizes frames.	FR-04
ECGClassifier	Runs SE-ResNet1D; returns probabilities and last-layer activations.	FR-06
EchoSegmenter	Runs the lightweight U-Net; returns per-frame LV masks and an embedding.	FR-07
VolumeEstimator	Builds the volume curve, finds ED/ES, computes EDV, ESV and EF (None if EDV ≤ 0).	FR-07
Uncertainty Estimator	Aggregates T passes / K members; computes spread, entropy and reduced-EF probabilities.	FR-14, FR-15
ExplanationGenerator	Produces ECG saliency images and LV overlays.	SRS Table 18
FusionModel	Combines ECG and Echo embeddings with EF features; returns fused probabilities and gate weights.	FR-18, FR-19
ReferralSummaryService	Validates the authenticated request, authorizes access to the selected case, retrieves available case and analysis data, coordinates summary generation and persistence, and records the operation.	UC-09, FR-42, FR-43, FR-46, FR-49
ReferralSummaryBuilder	Populates a predefined, deterministic and versioned summary template using available stored outputs. It does not invoke another model or alter the source results.	UC-09, FR-44, FR-45

Class	Responsibility	Traces to
ReferralSummaryRepository	Persists and retrieves ReferralSummary records through the Supabase data-access layer.	UC-09, FR-46, FR-47
ReferralSummary	Represents the structured summary content and metadata, including case reference, generating user, template version, analysis references, and generation timestamp.	UC-09, FR-46

6. Database Diagram

Figure 6 shows the physical PostgreSQL schema on Supabase. Supabase keeps auth.users; every application table is in the public schema, with RLS. Most cases have either one ECG result, one Echo result, one fusion result, or one triage decision (enforced with UNIQUE foreign keys) so the case record is the natural join point. BINARYs are never stored in rows tables contain only Storage paths (FR-36). Generate UUIDs for the primary keys of every table except audit_events which has an identity column for quick append-only inserts. All timestamps are timestamptz with default now().

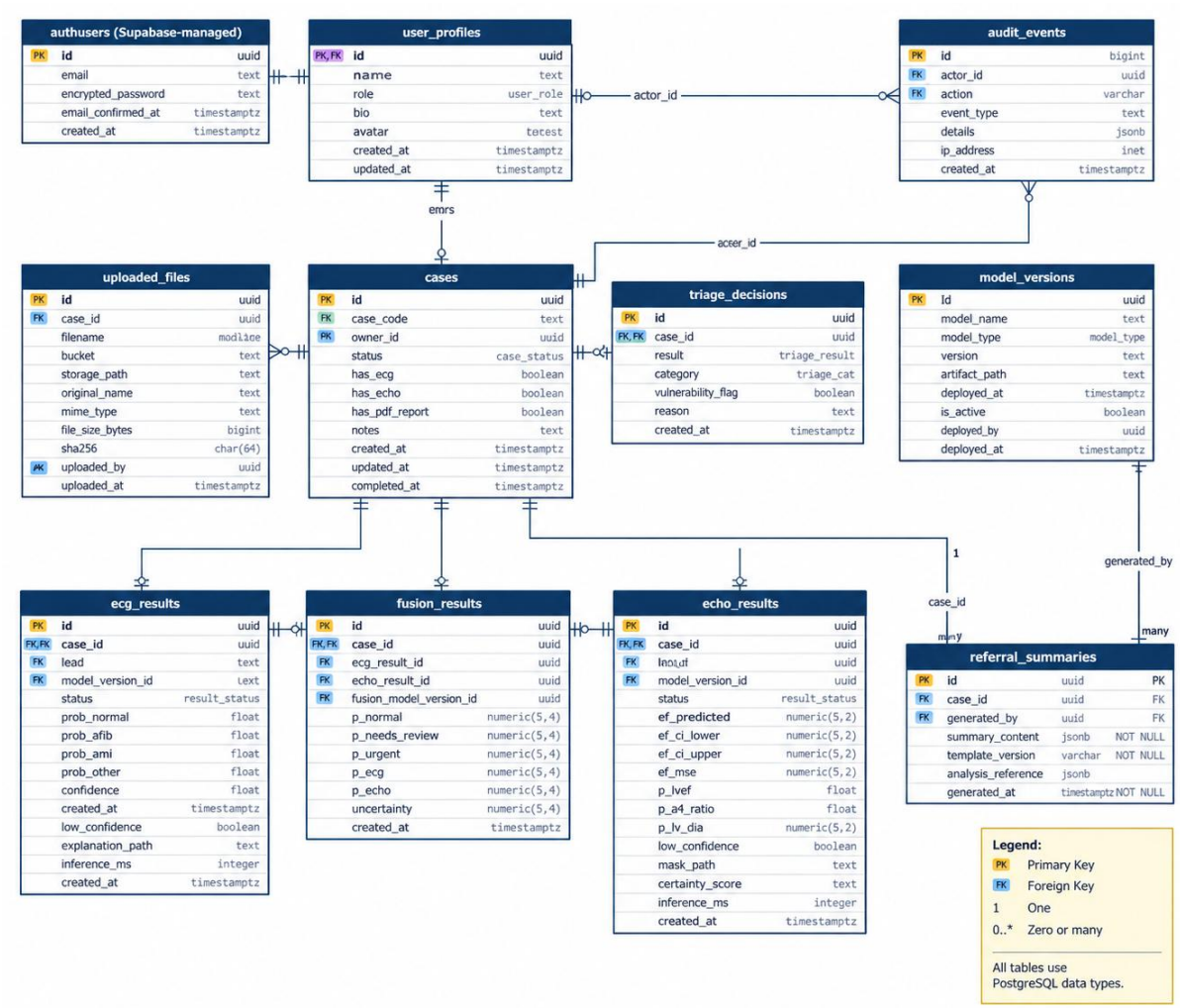


Figure 6: Database Diagram (Supabase PostgreSQL)

6.1. Data Dictionary

Table 11: Data Dictionary - user_profiles

Column	Type	Constraints	Description
id	uuid	PK; FK → auth.users(id) ON DELETE CASCADE	Same identifier as the Supabase Auth user.
full_name	text	NOT NULL	Display name.
role	user_role	NOT NULL DEFAULT 'clinician'	clinician | nurse | admin; changed only by the admin endpoint.
facility	text	NULL	Optional health facility name.
is_active	boolean	NOT NULL DEFAULT true	Deactivated users are rejected by AuthGuard.
created_at / updated_at	timestamptz	NOT NULL DEFAULT now()	Audit timestamps.

Table 12: Data Dictionary - cases

Column	Type	Constraints	Description
id	uuid	PK	Internal case identifier.
case_code	text	UNIQUE, NOT NULL	Pseudonymous code 'CA-YYYY-NNNNNN' generated from a sequence.
owner_id	uuid	FK → user_profiles(id), NOT NULL	User who submitted the case.
status	case_status	NOT NULL DEFAULT 'draft'	Lifecycle state (see the Enumerated Types table).
has_ecg / has_echo	boolean	NOT NULL DEFAULT false	Modalities present.
pair_eligible	boolean	NOT NULL DEFAULT false	True only when patient linkage and timing are verified.
notes	text	NULL, ≤ 500 chars	Free text without personal identifiers.
created_at / updated_at / completed_at	timestamptz	completed_at NULL until finished	Timestamps.

Table 13: Data Dictionary - uploaded_files

Column	Type	Constraints	Description
id	uuid	PK	File identifier.
case_id	uuid	FK → cases(id) ON DELETE CASCADE; UNIQUE (case_id, modality)	One file per modality per case.
modality	modality	NOT NULL	ecg | echo.
bucket / storage_path	text	NOT NULL; storage_path UNIQUE	Path '{owner}/{case}/{modality}/{file}' in case-inputs.

Column	Type	Constraints	Description
original_name / mime_type	text	NOT NULL	Client-supplied metadata, re-validated by the backend.
size_bytes	bigint	CHECK (size_bytes > 0)	Checked against the configured limit.
sha256	char(64)	NOT NULL	Integrity check of the stored object.
uploaded_by	uuid	FK → user_profiles(id)	Uploader.
uploaded_at	timestamptz	DEFAULT now()	Upload time.

Table 14: Data Dictionary - model_versions

Column	Type	Constraints	Description
id	uuid	PK	Model version identifier.
model_name	text	UNIQUE (model_name, version)	ecg_se_resnet1d | echo_lw_unet | gated_fusion.
modality	model_modality	NOT NULL	ecg | echo | fusion.
version / artifact_path	text	NOT NULL	Semantic version and path of the .onnx file in the image.
quantization	text	NOT NULL	fp32 | int8-dynamic | int8-static.
sha256	char(64)	NOT NULL	Artifact checksum verified at start-up.
is_active / deployed_at	boolean / timestamptz	Partial UNIQUE: one active per model_name	Currently served version.

Table 15: Data Dictionary - ecg_results

Column	Type	Constraints	Description
id	uuid	PK	Result identifier.
case_id	uuid	FK → cases(id), UNIQUE	At most one ECG result per case.
file_id / model_version_id	uuid	FK → uploaded_files / model_versions	Input file and model used.
status	result_status	NOT NULL	completed | failed.
predicted_class	ecg_class	NULL if failed	normal | arrhythmia | ischemic | other_abnormal.
prob_normal … prob_other	numeric(5,4)	CHECK 0–1	Mean class probabilities (sum ≈ 1).
confidence / entropy	numeric	CHECK ≥ 0	Max mean probability; predictive entropy.
low_confidence	boolean	NOT NULL	True when entropy > τ_ECG.
explanation_path	text	NULL	Saliency image in case-artifacts.
inference_ms / created_at	integer / timestamptz		Latency (benchmarking) and time.

Table 16: Data Dictionary - echo_results

Column	Type	Constraints	Description
id, case_id, file_id, model_version_id, status	uuid / result_status	As in ecg_results	Identification and status.
ed_frame / es_frame	integer	NULL if unavailable	Frames of maximum and minimum volume.
edv_ml / esv_ml	numeric(7,2)	NULL if unavailable; CHECK ≥ 0	Estimated volumes.
ef_percent	numeric(5,2)	NULL if unavailable; CHECK 0–100	EF = (EDV − ESV) / EDV × 100.
ef_std	numeric(5,2)	NULL	Spread of EF samples.
p_ef_lt50 / p_ef_le40	numeric(5,4)	CHECK 0–1	Reduced-EF probabilities.
low_confidence	boolean	NOT NULL	True when ef_std > τ_EF.
mask_path / overlay_path	text	NULL	Mask archive and overlay video in case-artifacts.
inference_ms / created_at	integer / timestamptz		Latency and time.

Table 17: Data Dictionary - fusion_results

Column	Type	Constraints	Description
id	uuid	PK	Result identifier.
case_id	uuid	FK → cases(id), UNIQUE	One fusion result per case.
ecg_result_id / echo_result_id	uuid	FK → ecg_results / echo_results	The paired inputs.
model_version_id	uuid	FK → model_versions	Fusion model used.
p_normal / p_needs_review / p_urgent	numeric(5,4)	CHECK 0–1	Fused triage probabilities.
gate_ecg / gate_echo	numeric(5,4)	CHECK 0–1; sum = 1	Modality reliance (FR-19).
uncertainty / created_at	numeric / timestamptz		Fused uncertainty and time.

Table 18: Data Dictionary - triage_decisions

Column	Type	Constraints	Description
id	uuid	PK	Decision identifier.
case_id	uuid	FK → cases(id), UNIQUE	One decision per case.
source	triage_source	NOT NULL	fusion | echo | ecg.
category	triage_cat	NOT NULL	normal | needs_review | urgent_referral.
uncertainty_flag	boolean	NOT NULL	True when an uncertainty override applied.

Column	Type	Constraints	Description
rationale	text	NOT NULL	Human-readable reason, e.g. 'EF 46.8% in 40–50% band'.
created_at	timestamptz	DEFAULT now()	Decision time.

Table 19: Data Dictionary - audit_events

Column	Type	Constraints	Description
id	bigint	PK, GENERATED ALWAYS AS IDENTITY	Event sequence.
case_id	uuid	NULL; FK → cases ON DELETE SET NULL	Related case, if any.
actor_id	uuid	NULL; FK → user_profiles	NULL means the system acted.
event_type	text	NOT NULL	CASE_CREATED, FILE_REGISTERED, ANALYSIS_QUEUED, ECG_ANALYSED, ECHO_ANALYSED, FUSION_RUN, TRIAGE_GENERATED, ROLE_CHANGED, USER_DEACTIVATED, ACCESS_DENIED.
status / details	text / json	details without personal data	Outcome and context.
created_at	timestamptz	DEFAULT now()	Event time; table is append-only (no UPDATE/DELETE grants).

Table 20: Data Dictionary - referral-summaries

Column	Type	Constraints	Description
id	uuid	PK	Unique identifier for the generated Referral Summary.
case_id	uuid	FK, NOT NULL	Identifies the case to which the summary belongs.
generated_by	uuid	FK → user_profiles(id), NOT NULL	Identifies the authenticated Clinician who requested generation.
summary_content	JSONB or TEXT	NOT NULL	Stores the structured summary content generated from available case information and analysis results.
template_version	VARCHAR	NOT NULL	Identifies the predefined summary-template version used.
analysis_reference	JSONB or actual FK fields	NOT NULL Where Applicable	Stores references to the completed analysis records used to generate the summary.
generated_at	timestamptz	NOT NULL DEFAULT now()	Records the date and time of summary generation.

Referral Summary records are stored in Supabase PostgreSQL and linked to their corresponding cases. The stored record contains the summary content and generation metadata. Supabase Storage is not required for the normal summary-generation workflow; it is used for a PDF summary only if PDF export is implemented.

6.2. Enumerated Types, Storage Buckets and Access Policies

Table 21: Enumerated Types

Type	Values
user_role	clinician, nurse, admin
case_status	draft, queued, processing, completed, partially_completed, failed
modality / model_modality	ecg, echo / ecg, echo, fusion
result_status	completed, failed
ecg_class	normal, arrhythmia, ischemic, other_abnormal
triage_source / triage_cat	fusion, echo, ecg / normal, needs_review, urgent_referral

Table 22: Supabase Storage Buckets

Bucket	Visibility	Path pattern	Access
case-inputs	Private	{user_id}/{case_id}/{ecg|echo}/{file}	INSERT/SELECT by the owner only — policy checks that the first folder equals auth.uid(); backend reads with service role after authorize().
case-artifacts	Private	{user_id}/{case_id}/{artifact}	No client policy; written by the backend and read through 60-second signed URLs.

Table 23: Row-Level Security Policies

Table	Authenticated client (browser)	Backend (service role)
user_profiles	SELECT own row; UPDATE full_name and facility only (role and is_active column privileges revoked).	Full access through AdminRouter after role check.
cases	SELECT where owner_id = auth.uid().	INSERT/UPDATE after authorize().
uploaded_files	SELECT where the parent case is owned by auth.uid().	INSERT after validation.
ecg_results, echo_results, fusion_results, triage_decisions	SELECT where the parent case is owned by auth.uid(); no writes.	INSERT by Inference Orchestrator.
model_versions	SELECT (non-sensitive metadata).	INSERT/UPDATE at deployment.
audit_events	No access.	INSERT only; SELECT for admins via API.

Table	Authenticated client (browser)	Backend (service role)
referral_summaries	SELECT	An authenticated, active Clinician may retrieve a Referral Summary only when the Clinician is authorized to access its associated case. Access is restricted through the applicable Row-Level Security policy and backend authorization checks.
referral_summaries	INSERT	Summary insertion is performed through the authorized backend service after the clinician's JWT and case permissions have been validated.
referral_summaries	UPDATE / DELETE	Not permitted through ordinary client access unless an explicitly approved administrative or retention operation requires it.

7. Entity Relationship Diagram (ERD)

Figure 7 depicts the conceptual ERD in the Chen representation. The entities are defined as rectangles, relationships as diamonds and representative attributes as ellipses. Key attributes are underlined and multi-valued attributes (class probabilities, gate weights, fusion probabilities) are using double ellipses. EF is shown as a derived attribute because EF is calculated from EDV and ESV. It is the conceptual source of the physical schema found in Section 6: one-to-many relationships are foreign keys on the "many" side, and one-to-one relationships become UNIQUE foreign keys.

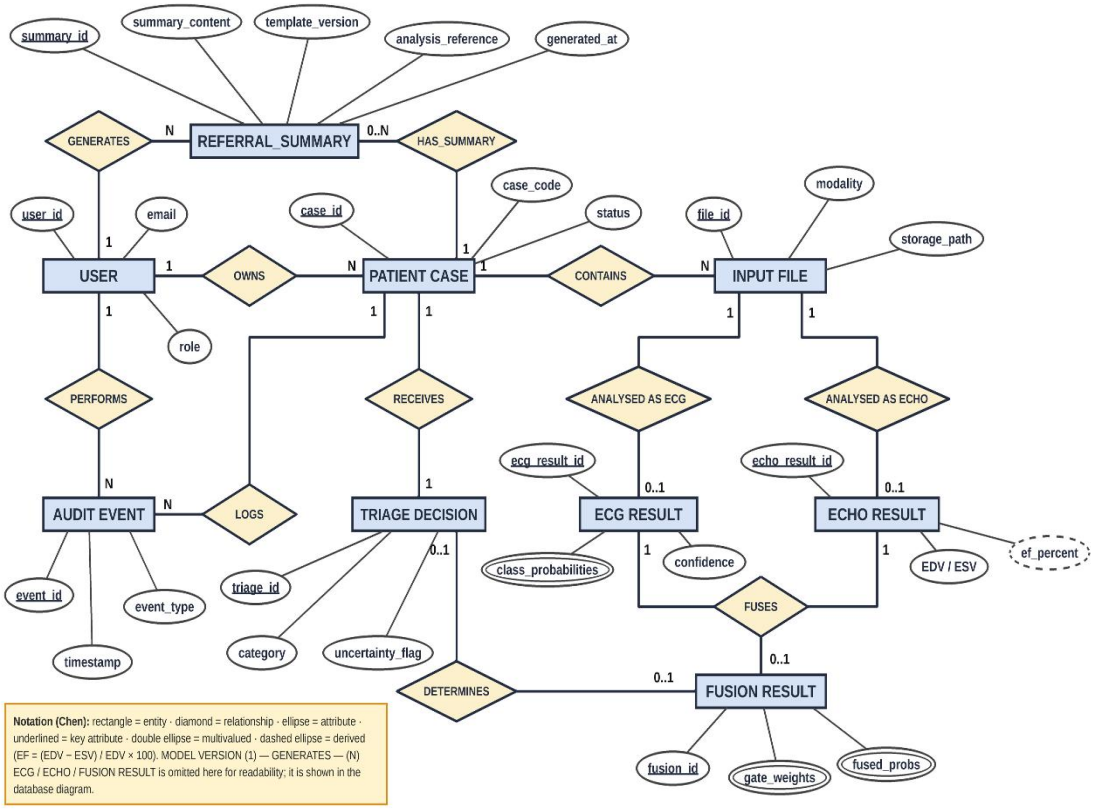


Figure 7: Entity Relationship Diagram (Chen Notation)

The ERD includes the REFERRAL_SUMMARY entity to represent summaries generated on explicit clinician request. Each summary is associated with one patient case and the user who initiated generation. The summary record stores its structured content, template version, generation timestamp, and references to the available analysis results. This relationship preserves the link between the summary and the original screening records without creating a separate diagnosis or changing the stored predictions.

Table 24: Relationships and Cardinalities

Relationship	Entities	Cardinality	Implementation
OWNS	User — Patient Case	1 : N	cases.owner_id → user_profiles.id
CONTAINS	Patient Case — Input File	1 : N (max 2, one per modality)	uploaded_files.case_id; UNIQUE (case_id, modality)

Relationship	Entities	Cardinality	Implementation
ANALYSED AS ECG	Input File — ECG Result	1 : 0..1	ecg_results.file_id
ANALYSED AS ECHO	Input File — Echo Result	1 : 0..1	echo_results.file_id
FUSES (ternary)	ECG Result, Echo Result — Fusion Result	1 : 1 : 0..1	fusion_results.ecg_result_id, echo_result_id
RECEIVES	Patient Case — Triage Decision	1 : 0..1	triage_decisions.case_id UNIQUE
DETERMINES	Fusion Result — Triage Decision	0..1 : 0..1	Shared case_id; source = fusion
PERFORMS	User — Audit Event	1 : N (optional)	audit_events.actor_id (NULL = system)
LOGS	Patient Case — Audit Event	1 : N	audit_events.case_id
GENERATES	Model Version — ECG / Echo / Fusion Result	1 : N each	model_version_id on each result table
PatientCase - ReferralSummary	Patient Case – Referral Summary	1:N	referral_summaries.case_id → cases.id
User - ReferralSummary	User – Referral Summary	1:N	`referral_summaries.generated_by` → `user_profiles.id` (or the actual user table used)

A case may have multiple Referral Summary records because a clinician may request a new summary after reviewing the case again. Each generated record remains associated with its originating case and the clinician who initiated the request. The records retain their generation timestamps and template versions.

8. Sequence Diagram

Below is the user flow for each use case of describing parts in the SRS. The unexpected number of sequence diagrams might be disconcerting but it is purely the result of the React component, FastAPI subcomponent, ONNX Runtime, and Supabase service interacting with each other at run time. Solid arrows represent calls, dashed arrows represent returns, and alt, opt, and loop fragments indicates an alternative, optional, and repeated behaviours. The Test Case Message Numbers are specified in the test cases of Section 10.

8.1. Register User (UC-01)

Supabase Auth is used to register a new Clinician/Nurse. It fires a database trigger to create user profile with default role set to clinician. This stops users from gaining Admin access by registering from the public-facing site (FR-32). Only after the user has confirmed their email via the verification link is access to any of the protected features allowed (FR-31).

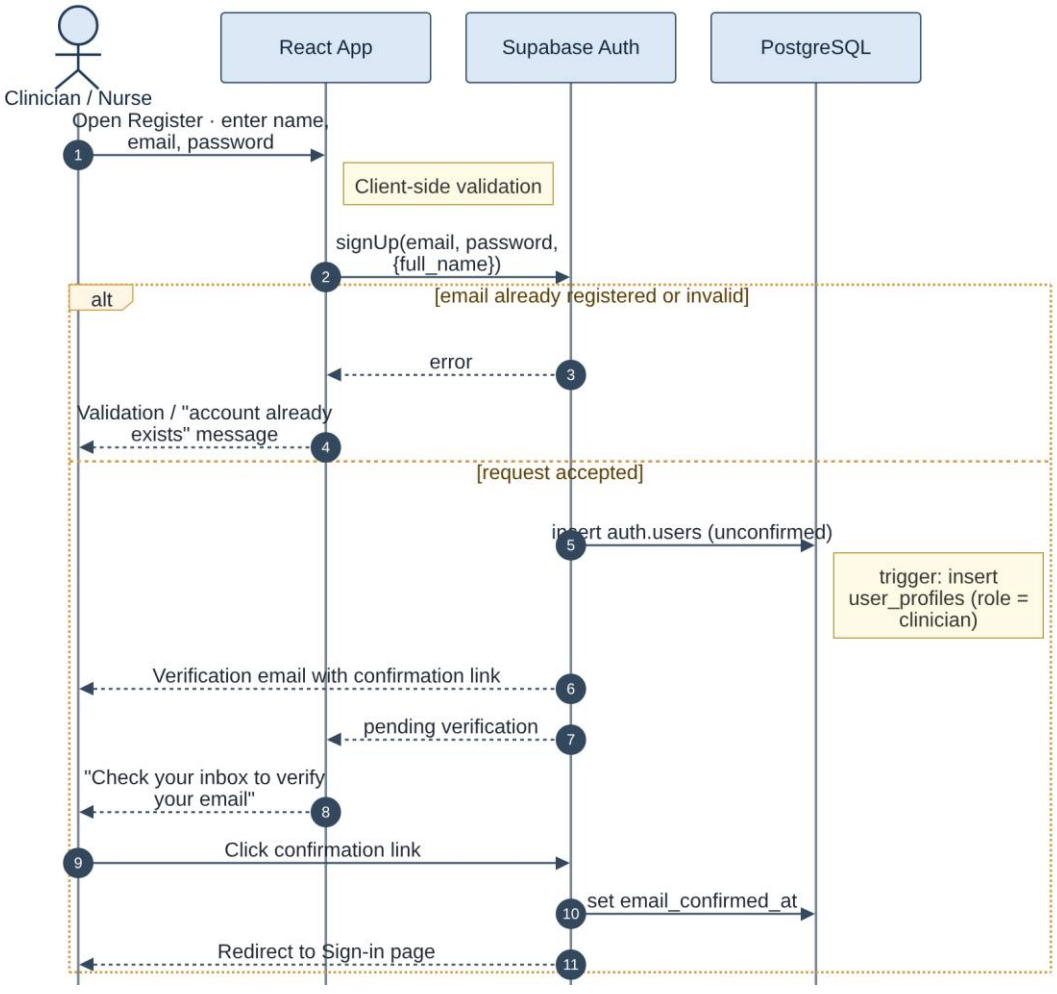


Figure 8: Sequence Diagram - Register User (UC-01)

8.2. Authenticate User (UC-02)

The credentials are confirmed by Supabase, which in turn sends back a JWT session. Then frontend will call GET /api/v1/me; AuthGuard checks the token and profile status before landing on the role-specific dashboard. Signing out and password recovery flows are all performed via the Supabase client.

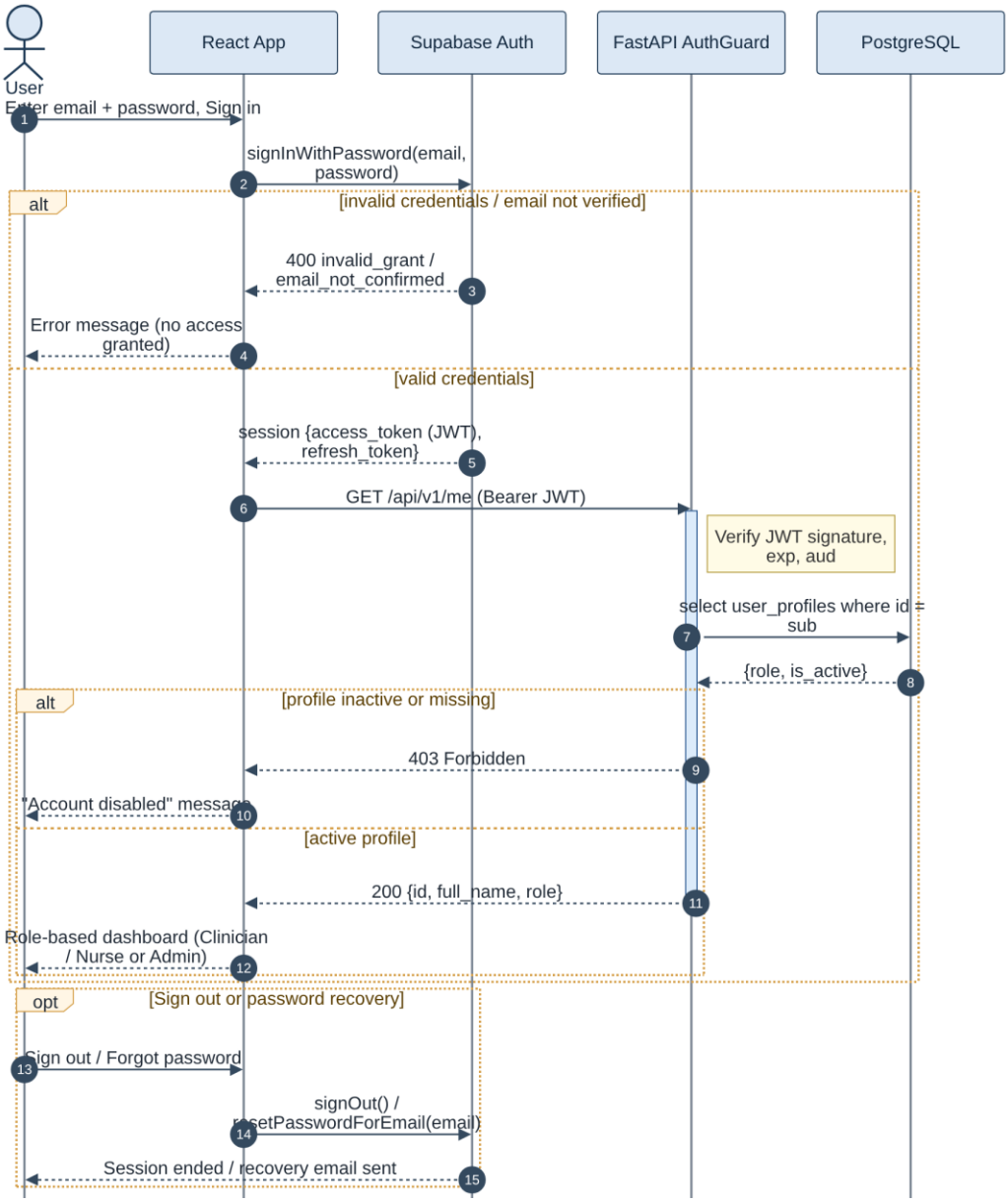


Figure 9: Sequence Diagram - Authenticate User (UC-02)

8.3. Submit Patient Case (UC-03)

A draft case is first made, each selected file is uploaded first direct to the private case-inputs bucket into user folder, and metadata registered and validated by the backend. Queues a background task on submission and gives back 202. If an upload fails, the case remains a draft and no success message is shown (UC-03 alternative 3a).

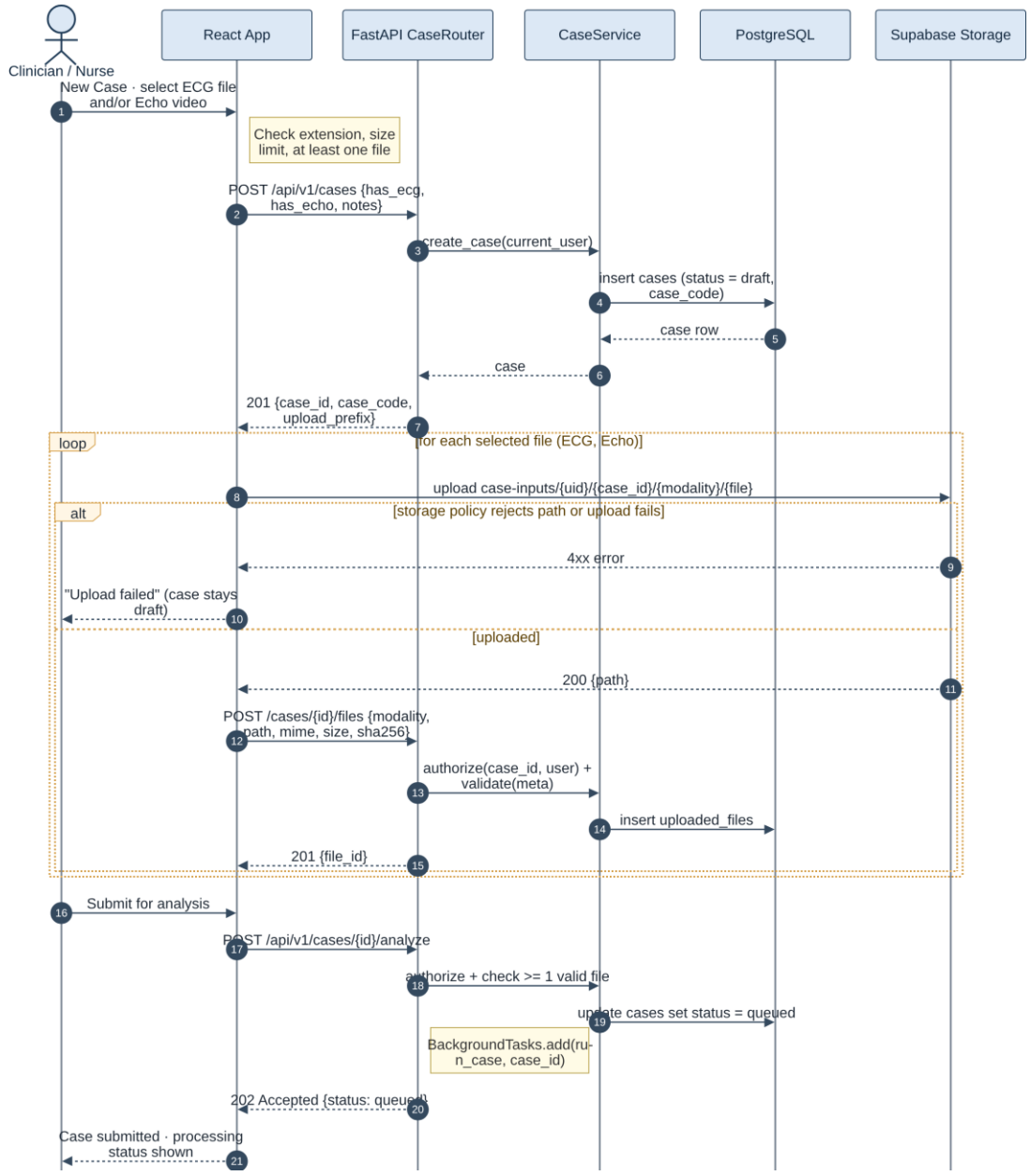


Figure 10: Sequence Diagram - Submit Patient Case (UC-03)

8.4. Analyze ECG (UC-04)

The orchestrator now downloads the ECG, and it, for compatbile parts only, harmonizes them to 12 leads at 250 Hz. It performs T stochastic passes (or K ensemble members) through SE-ResNet1DOMNX, aggregates uncertainty, and generates the Grad-CAM style saliency explanation if available. Results, the low-confidence flag and the artifact path are stored. An unreadable file leads to a failed result rather than prediction.

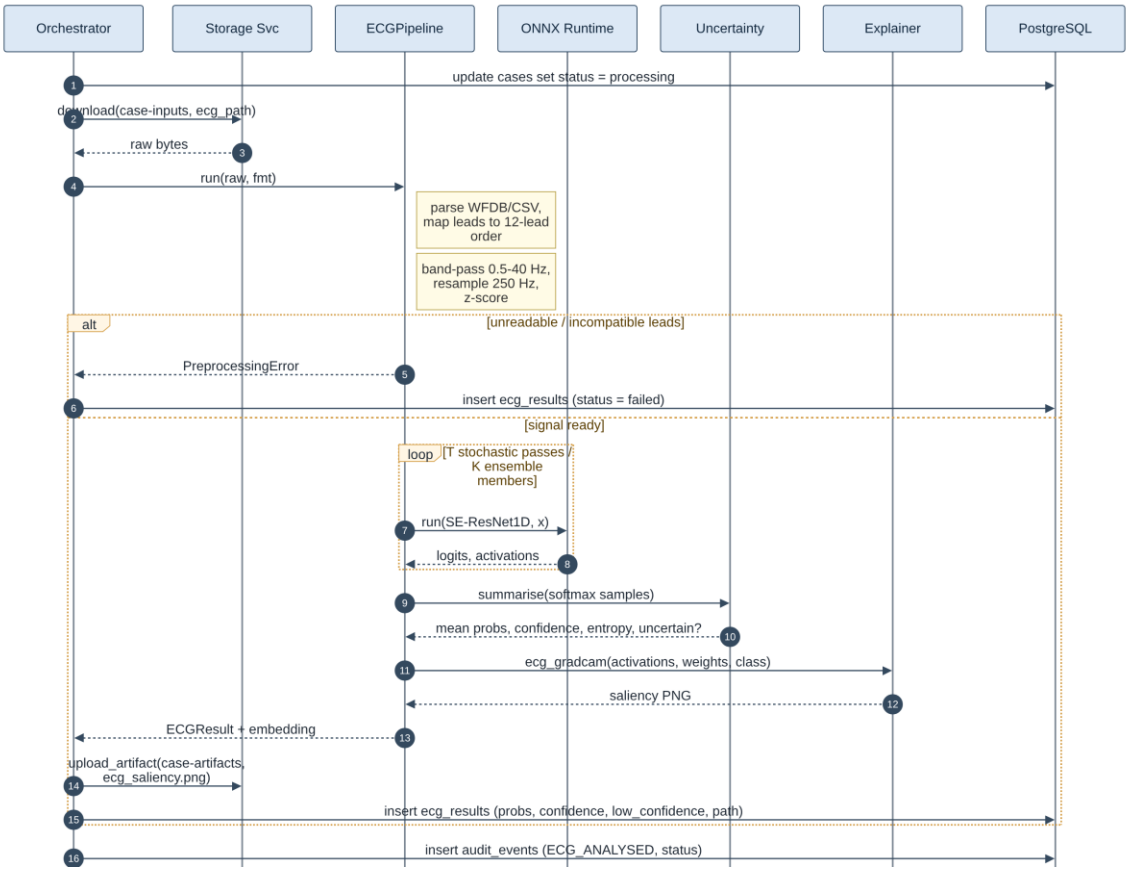


Figure 11: Sequence Diagram - Analyze ECG (UC-04)

8.5. Analyze Echocardiogram (UC-05)

The frames are decoded, normalized, then passed to a small U-Net for segmentation in each frame, VolumeEstimator is returned EDV, ESV, and EF for each stochastic pass. In case of segmentation failure or non-positivity of EDV the EF is marked unavailable, not estimated (UC-05 alternative 3a).

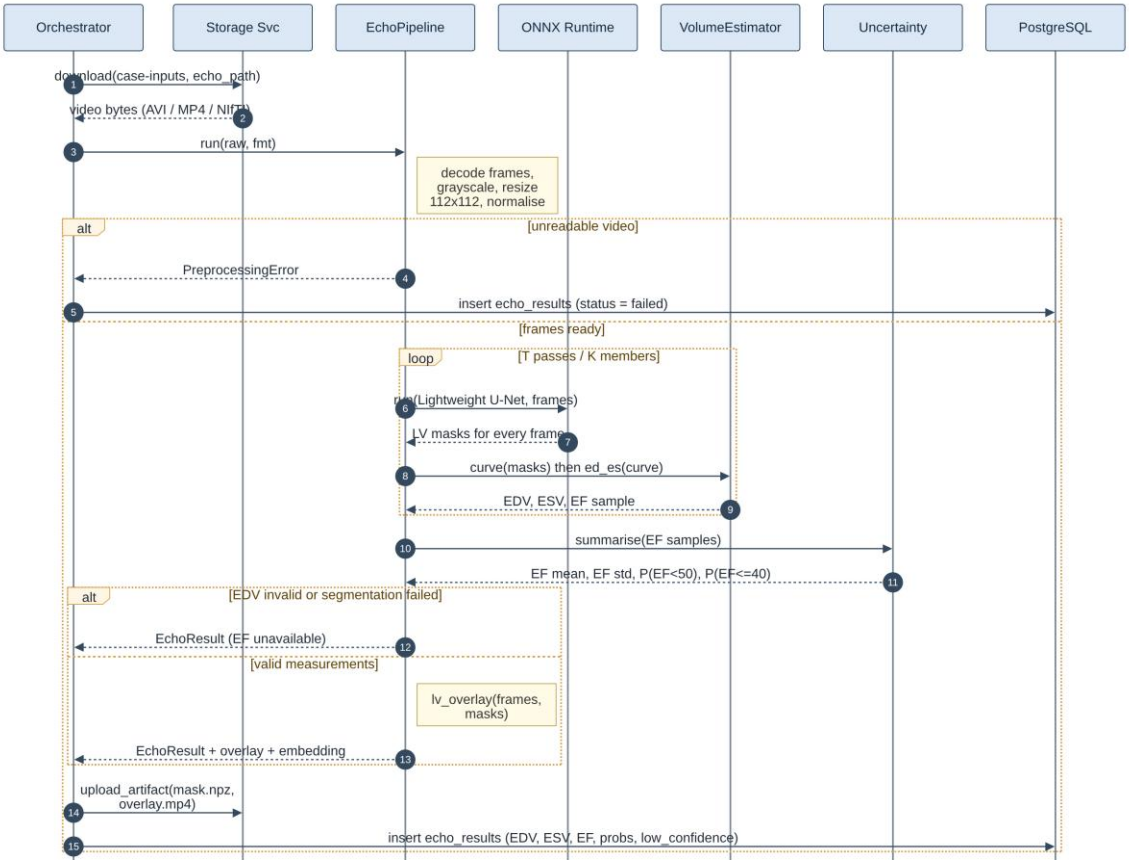


Figure 12: Sequence Diagram - Analyze Echocardiogram (UC-05)

8.6. Generate Triage Decision (UC-06)

Verifiable pairs through gated fusion model these forward the fused probabilities and gated weights; all other exchange with Echo EF bands, with the uncertainty override to Needs Review. No triage disposition assigned for only- ECG cases until an ECG-only policy is set (OI-2). Every branch updates the case status an audit event.

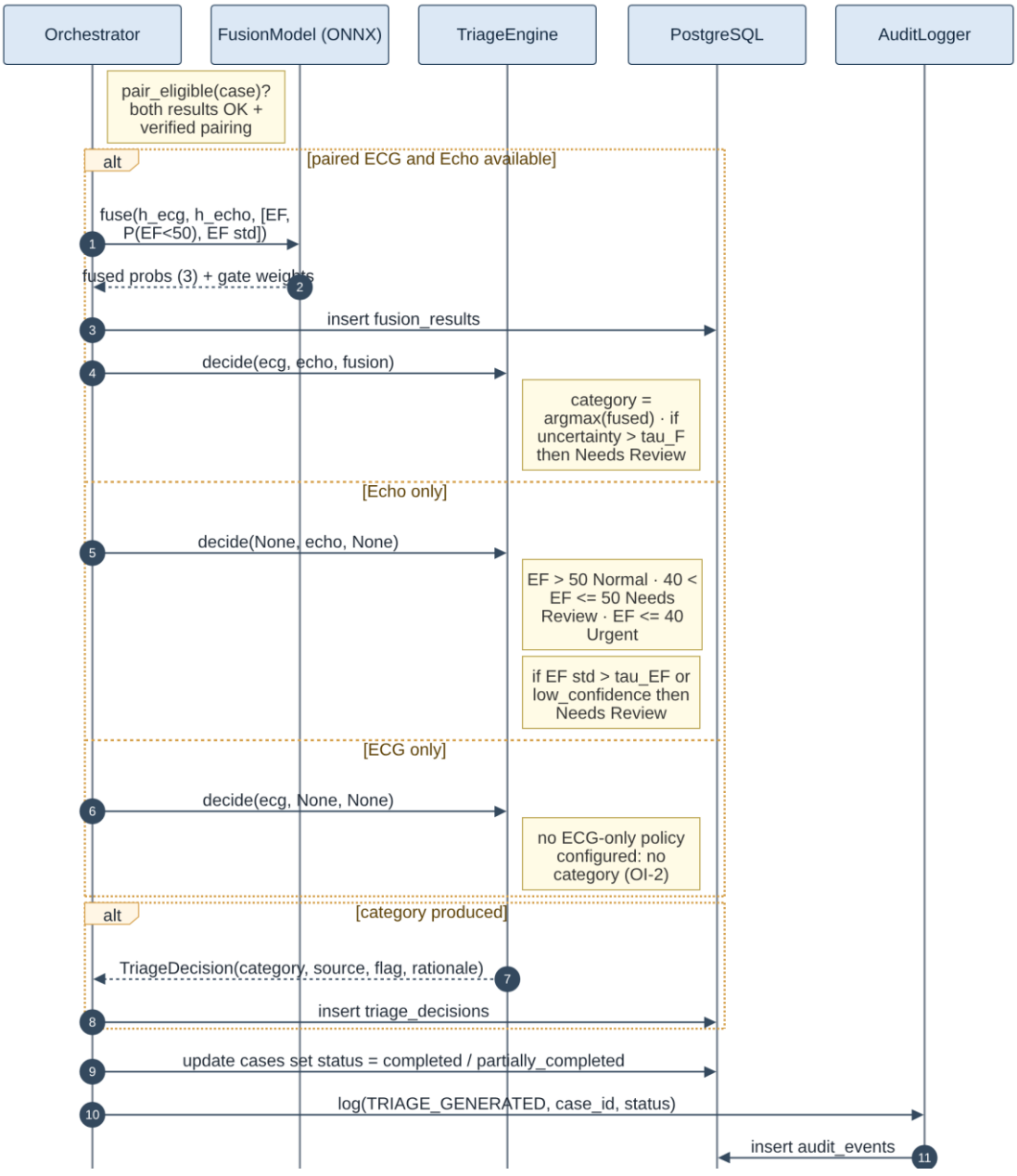


Figure 13: Sequence Diagram - Generate Triage Decision (UC-06)

8.7. View Result and Review Case Log (UC-07, UC-08)

The broker is listed by its cases that he places. When you open a case start an authorization flow; a still in progress case shows its status and is pollable; a completed case returns all results and a temporary signed URL for the saliency plot and LV overlay by default. View never overrides stored results.

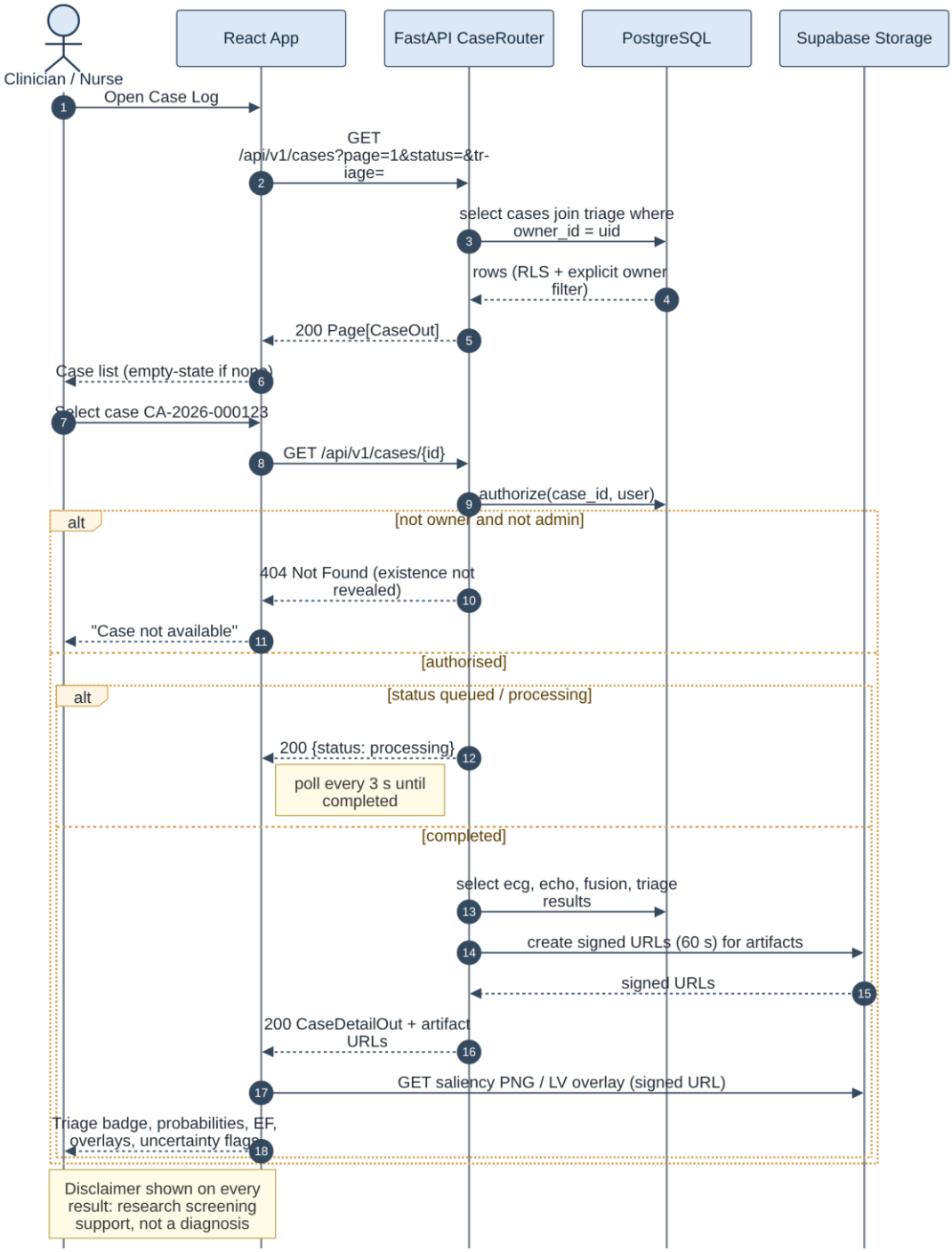


Figure 14: Sequence Diagram - View Result and Review Case Log (UC-07, UC-08)

8.8. Generate Referral Summary (UC-09)

Clinician triggers the Referral Summary Generation through the case-result screen after eyeballing the eligible screening results with good reference data available. The React front end will authenticate a POST to the FastAPI back end with the chosen case reference and the Supabase JWT of the user. AuthGuard looks after the authentication token, while ReferralSummaryService relies on CaseService to confirm the Clinician has the right to access the chosen case. Once validated, the service will jump into the Supabase PostgreSQL, thus accessing the accessible pseudo case info and the completed ECG, Echo, fusion, and triage DB get records, where needed. Then the referral summary content, organized according to the pre-designed deterministic template by ReferralSummaryBuilder, clearly flags missing modalities or nonexistence analysis outputs in place of fakery. The generated summary is stored through ReferralSummaryRepository, the related operation has been logged and the data is returned back to the React frontend for the clinician to confirm. Authentication, authorization, data gathering, rendering, or persistence failure will cause the system to return an appropriate error code and never indicate that a summary had been generated. The original analysis results and triage result are not affected.

Figure 15 — Sequence Diagram: Generate Referral Summary (UC-09)

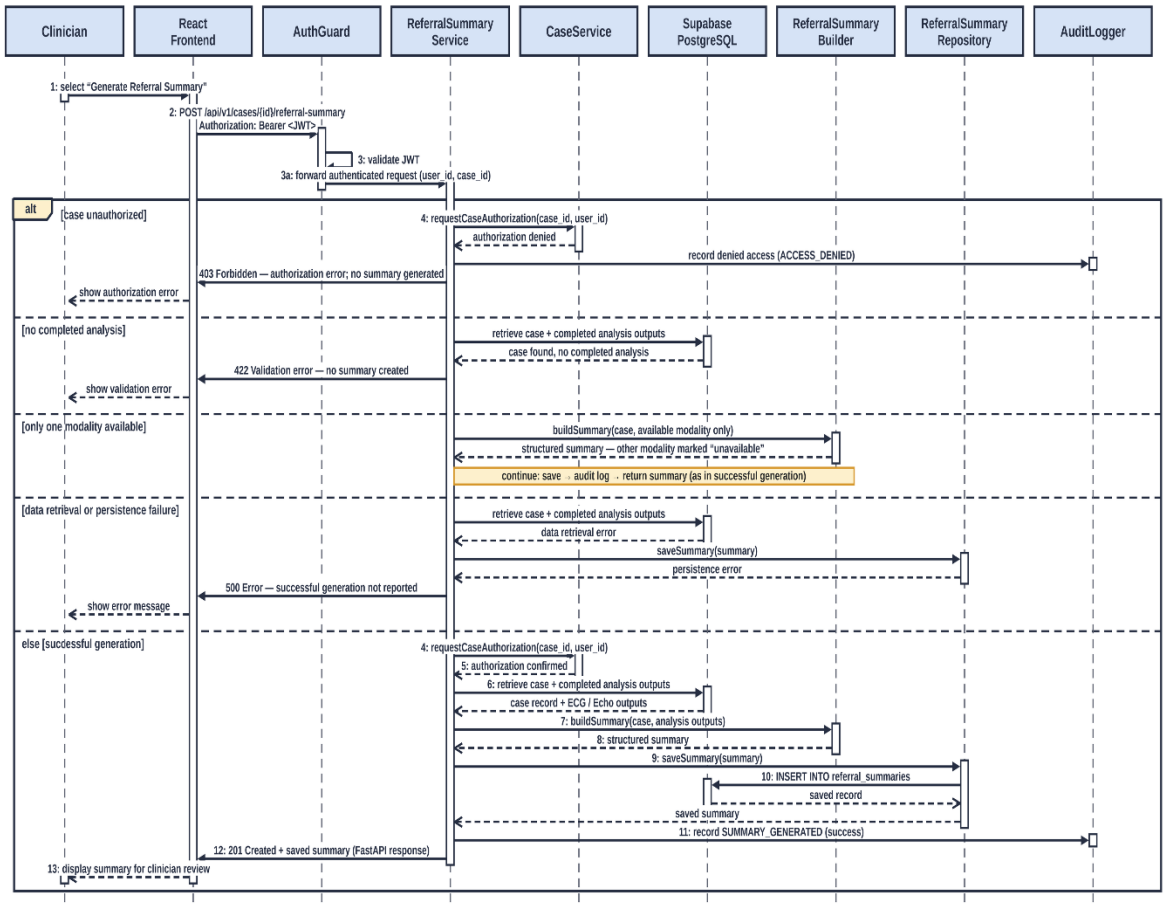


Figure 15: Sequence Diagram - Generate Referral Summary (UC-09)

9. System Interface Design

The user interface is designed for non-specialist healthcare workers using a laptop or desktop browser in a busy clinic. The following principles apply to every screen:

- **Plain-language results**: each result leads with the screening category and a one-line reason before any numbers (SRS 2.7, user constraints).
- **Colour is never the only signal**: Triage badges combine colour with text (green *Normal*, amber *Needs Review*, red *Urgent Referral*, grey *No category*), and low-confidence results carry a ⚑ flag and the words “Low confidence”.
- **Honest uncertainty**: Confidence values, EF spread and reduced-EF probabilities are shown next to the result; unavailable outputs are labelled as unavailable.
- **Persistent disclaimer**: Every page that shows results states that the output is screening support, not a diagnosis (SRS 5.2).
- **Privacy by design:** Only pseudonymous case codes are shown; the notes field warns against names, CNIC or phone numbers.
- **Consistency**: One navigation sidebar, a top bar with the user’s name and role, and the same badge, table and card components on every screen (React + Tailwind component library).

9.1. Sign-in Screen

Email/password sign-in with links to registration and password recovery (UC-02, FR-31). Failed attempts show a generic message that does not reveal whether the email exists. The left panel explains the purpose of the tool and the three screening categories.

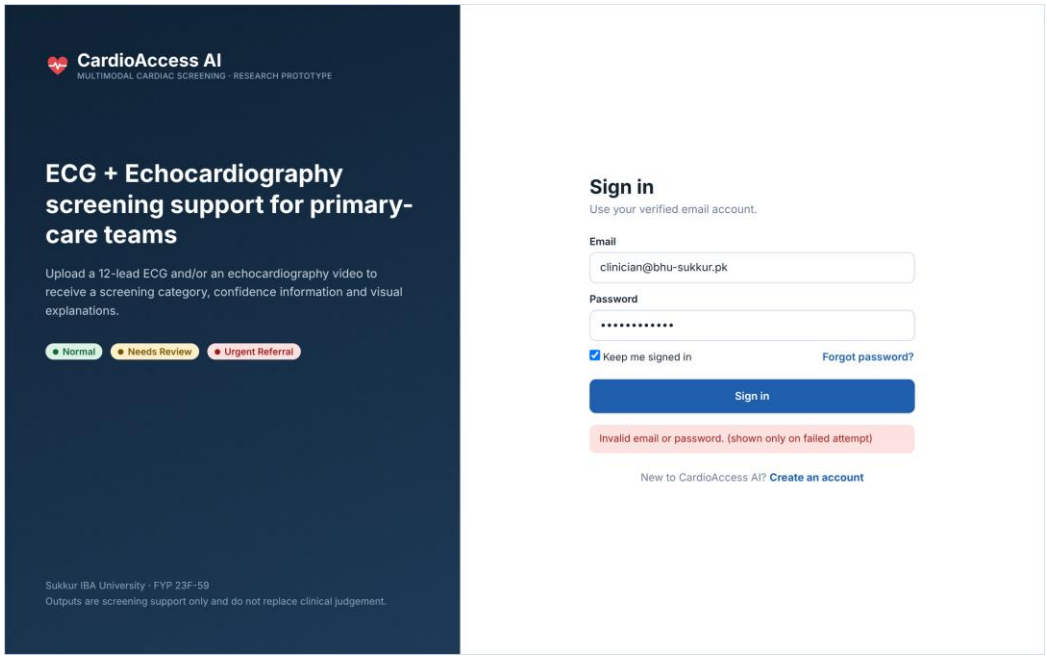


Figure 16: Interface - Sign-in Screen

9.2. Registration Screen

Collects name, work email, password (with live policy checks) and an optional facility, and requires the user to acknowledge the research-prototype status. The side panel explains email verification and that new accounts receive the Clinician/Nurse role (UC-01, FR-30, FR-32).

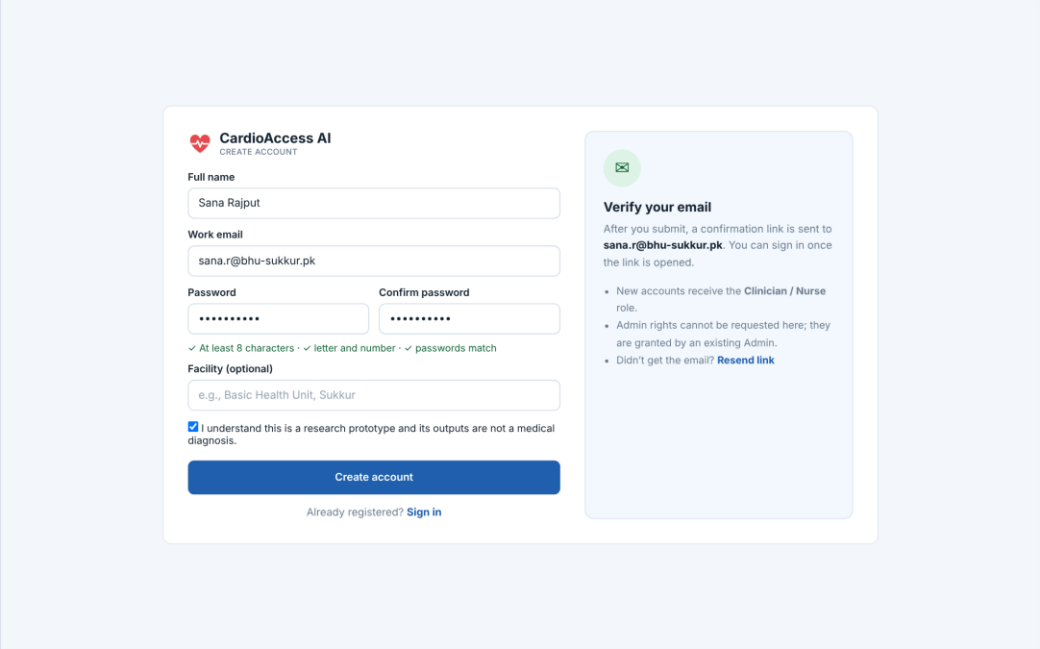


Figure 17: Interface - Registration Screen

9.3. Clinician Dashboard

Summary cards for cases this month, Needs Review, Urgent Referral and Processing; a recent-cases table with triage and status badges; and the triage mix for the last 30 days. The primary action, New Case, is always visible.

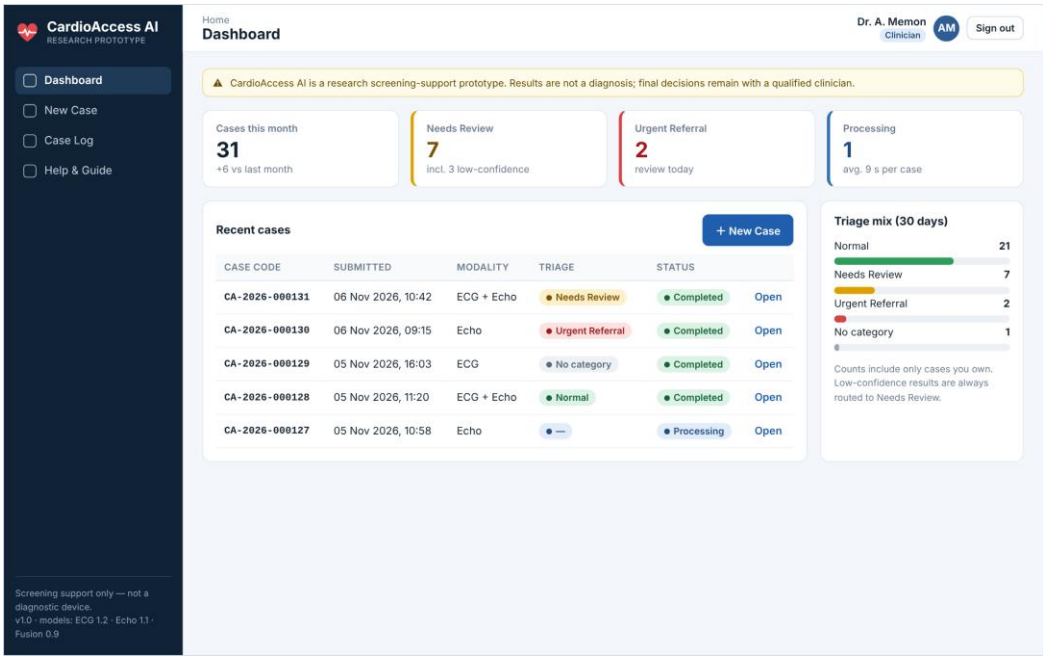


Figure 18: Interface - Clinician Dashboard

9.4. New Case Screen

Two independent upload areas for the ECG recording and the Echo video with accepted formats and size limits, upload progress, an automatically generated pseudonymous case code and a review panel. Submit is enabled only when at least one valid file has finished uploading (UC-03, FR-23).

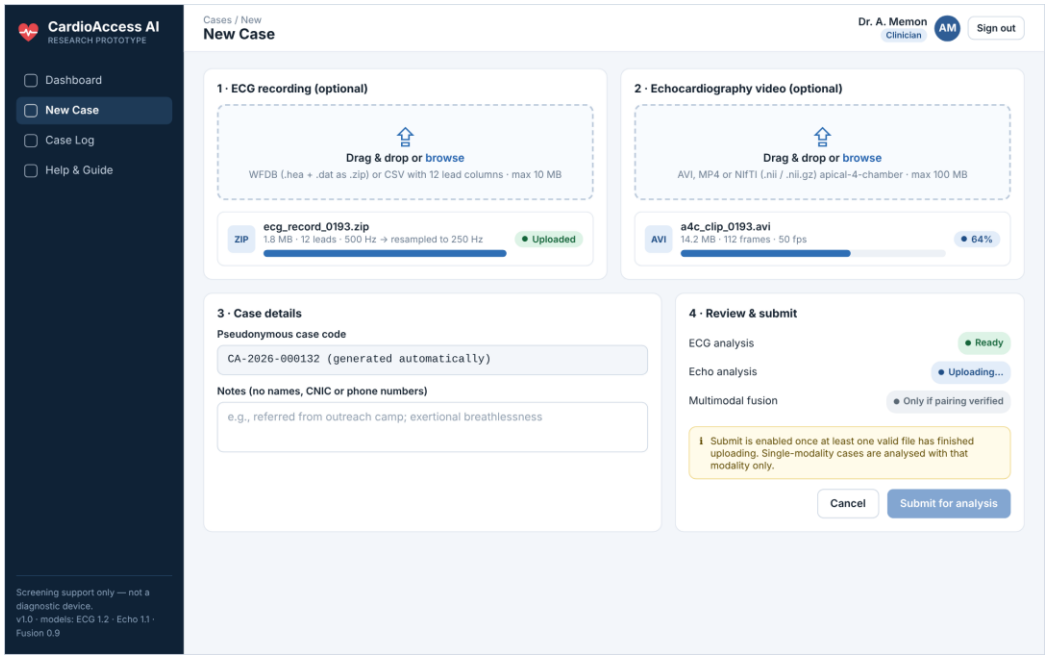


Figure 19: Interface - New Case Screen

9.5. Case Result and Explanation Screen

A triage banner with the fused probabilities, modality reliance and low-confidence flags; the ECG panel with the lead-selectable Grad-CAM-style saliency and the four class probabilities; the Echo panel with the LV overlay, EF ± spread, reduced-EF probability, EDV/ESV and the volume curve; and the processing timeline (UC-07, FR-19, FR-25, FR-26).

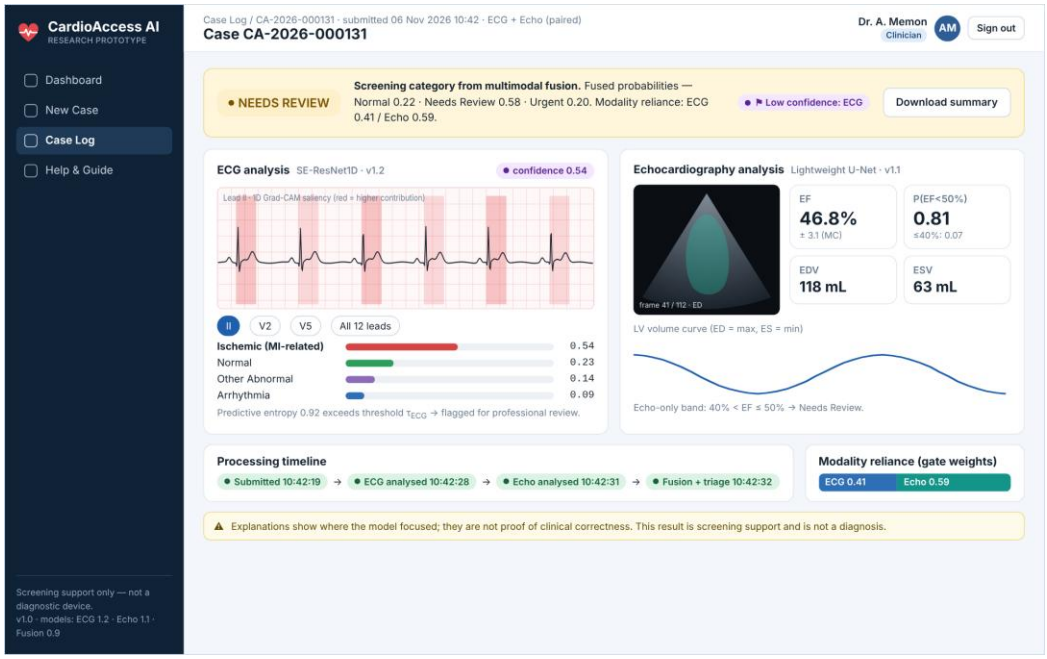


Figure 20: Interface - Case Result and Explanation Screen

The Case Result and Explanation screen displays all available screening outputs: triage category, ECG class probabilities, confidence and uncertainty information, echocardiographic measurements (where relevant), multimodal fusion results where applicable, and the respective explainability visualizations. Additionally, the screen provides the Generate Referral Summary action but only for an authenticated, active, and authorized Clinician. The action is only available when at least one relevant analysis has completed successfully. When the Clinician selects the action, React initiates an authenticated request to the FastAPI backend. Once it has successfully validated the authentication token and this case is allowed to be accessed, the backend fetches all case information as well as analysis outputs that have completed, creates the structured summary using a predefined template, and then stores it in Supabase PostgreSQL. React receives the stored summary result and has it rendered in a Summary Review panel in read-only mode with the following fields included: summary content, on which the summary was generated, any available modalities, and any unavailable information. It also makes clear that the summary is for clinician review and is not a diagnosis. The act of generating the summary does not change or affect the screening results, confidence or uncertainty values, analysis records, or triage outcome.

9.6. Case Log Screen

Searchable, filterable and paginated history of the user's own cases with modality, triage, processing status and low-confidence indicators (UC-08, FR-29). Failed analyses are shown explicitly.

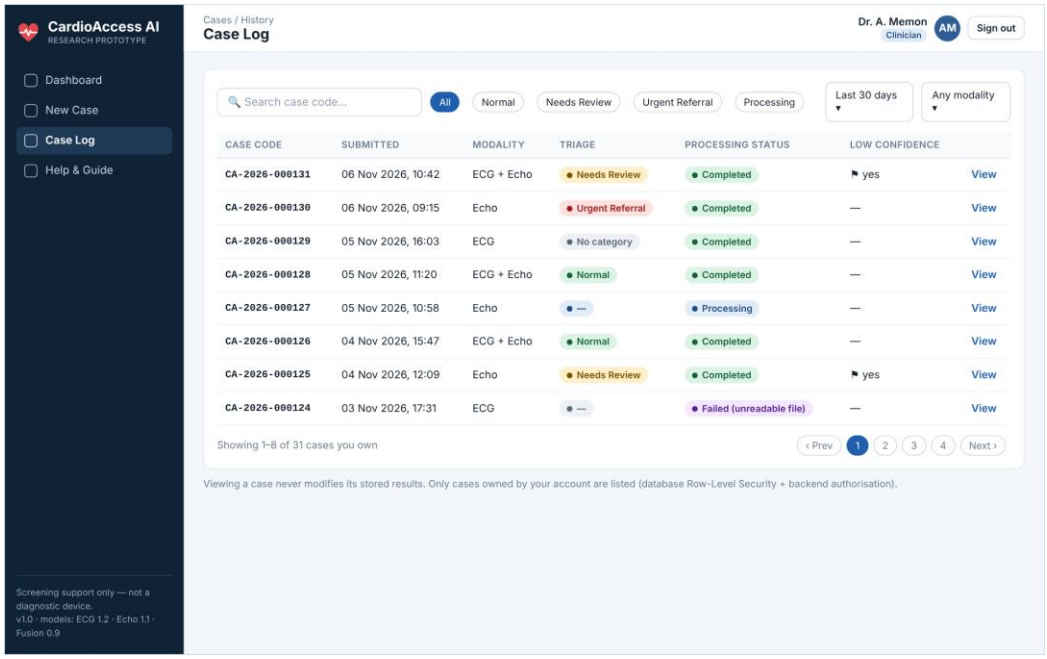


Figure 21: Interface - Case Log Screen

9.7. Admin - Users and Roles Screen

Lists registered users with role selector, verification/activation status and last sign-in, and shows the audit trail. Role and status changes are performed by the backend and audited (FR-32). The exact Admin capabilities remain subject to OI-5.

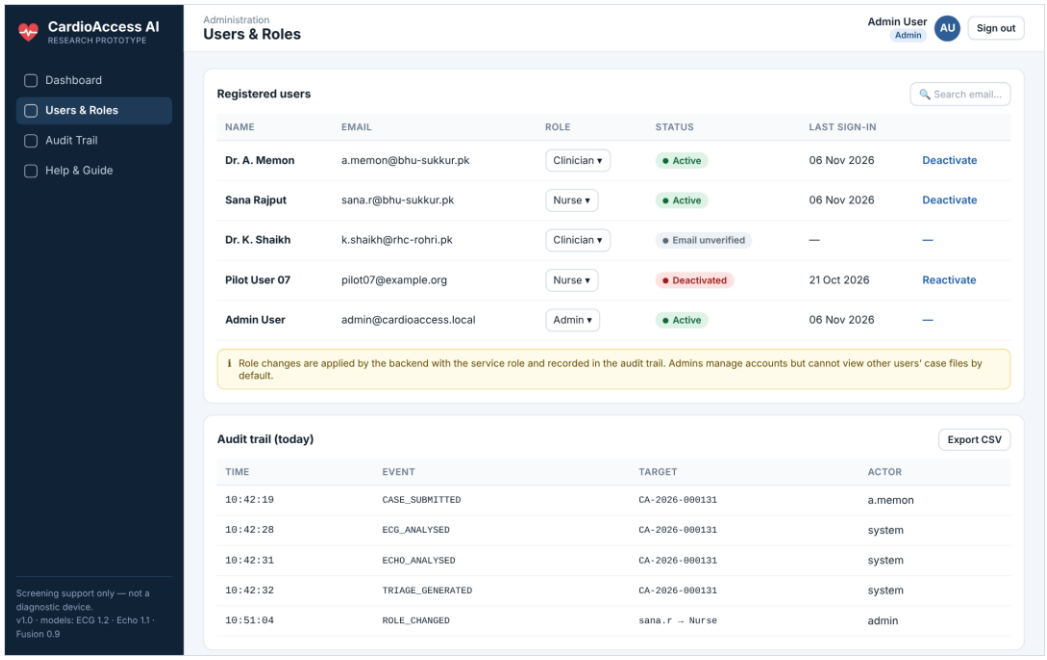


Figure 22: Interface (Admin) - Users and Roles Screen

Table 25: Screen-to-Requirement Traceability

Screen	Use cases	Functional requirements
Sign-in	UC-02	FR-30, FR-31, FR-33
Registration	UC-01	FR-30, FR-31, FR-32
Dashboard	UC-07, UC-08	FR-25, FR-26, FR-29
New Case	UC-03	FR-23, FR-34, FR-35
Case Result	UC-04 – UC-07	FR-16, FR-19, FR-24, FR-25, FR-26
Case Log	UC-08	FR-27, FR-29
Admin	UC-02 (Admin role)	FR-32, FR-33; SRS 5.3 audit trail
Case Result and Explanation	UC-07, UC-09	FR-41 to FR-48

10. Test Cases

The testing conducted uses automated unit and integration tests (back-end and pipeline tests are done using pytest; component testing for React environment). It also uses API level security tests, model-evaluation tests on held out data by using their simulations and a formative usability pilot. The tests point to a dedicated Supabase test project with the following seeds; two Services provided by Clinicians, one Nurse and an Admin, and with de-identified ECG and Echo files. The Reference External (Ex. the SRS UC) and functional requirements (FR). Status for all test cases is Pending in awaiting implementation.

Table 26: Test Cases - Authentication and Access Control

ID	Test case	Pre-condition	Steps / input	Expected result	Ref.
TC-01	Register with valid details	App reachable; email not registered	Open Register; enter name, valid email, compliant password; submit	Account created unconfirmed; verification email sent; “check inbox” shown; user_profiles row with role clinician	UC-01, FR-30
TC-02	Register with existing email	Email already registered	Submit registration with that email	“Account already exists” message; no duplicate account	UC-01 3a
TC-03	Register with invalid input	—	Blank email; weak password; mismatched confirmation	Inline validation messages; no request sent to Supabase	UC-01 2a
TC-04	Sign in before verification	Unverified account	Sign in with correct credentials	Access denied with “verify your email”; no session	UC-02 2b, FR-31
TC-05	Sign in with valid credentials	Verified, active account	Sign in	JWT session; GET /me returns 200; role-specific dashboard shown	UC-02, FR-33
TC-06	Sign in with wrong password	Verified account	Enter wrong password	Generic error; no token issued	UC-02 2a
TC-07	Password recovery	Verified account	Forgot password → open email link → set new password	Recovery email received; new password works; old one fails	FR-31
TC-08	Sign out	Signed in	Click Sign out; open /dashboard	Session cleared; redirected to Sign-in	UC-02 2d
TC-09	Admin role cannot be self-assigned	Signed-in clinician	Sign up with metadata role=admin; update own role via supabase-js	Profile role stays clinician; update rejected by RLS/column privileges	FR-32, FR-37
TC-10	Admin changes a user’s role	Admin signed in	PATCH /admin/users/{uid} {role: nurse}	200; role updated; ROLE_CHANGED audit event	FR-32
TC-11	Clinician calls admin endpoint	Clinician token	GET /api/v1/admin/users	403 Forbidden	FR-32, FR-33

Table 27: Test Cases - Case Submission and Storage

ID	Test case	Pre-condition	Steps / input	Expected result	Ref.
TC-12	Submit ECG-only case	Signed in; valid WFDB .zip	New Case → upload ECG → Submit	cases + uploaded_files rows; status queued; 202 Accepted; dashboard shows Processing	UC-03, FR-23, FR-34
TC-13	Submit ECG + Echo case	Valid ECG and AVI	Upload both → Submit	Two file rows; has_ecg and has_echo true; both lanes run	UC-03
TC-14	Unsupported file type	—	Upload a .docx as ECG	Rejected on client and by API (415); no file row	UC-03 2a, FR-23
TC-15	Oversized file	Limit configured	Upload a 150 MB video	Rejected with size message (413)	FR-23, OI-1
TC-16	Submit without a valid file	Draft case, no files	Press Submit / call /analyze	Button disabled; API returns 422	UC-03 2c
TC-17	Upload interrupted	Network dropped mid-upload	Upload Echo video	Error shown; case stays draft; no success message	UC-03 3a
TC-18	Upload into another user's folder	User A token	Upload to case-inputs/{userB}/…	Storage policy denies (403)	FR-35, SRS 5.3

Table 28: Test Cases - ECG and Echo Analysis

ID	Test case	Pre-condition	Steps / input	Expected result	Ref.
TC-19	ECG inference on valid record	Valid 12-lead, 500 Hz	Run analysis	Signal resampled to 250 Hz; 4 probabilities sum to 1 ± 0.001; class = arg-max; saliency stored; ecg_results row	UC-04, FR-40
TC-20	Unreadable ECG file	Corrupted .zip	Run analysis	ecg_results status failed; no probabilities displayed; audit event written	UC-04 2a
TC-21	Low-confidence ECG	Input with entropy > τ_ECG	Run analysis	low_confidence = true; ⚑ flag shown in UI	UC-04 4a, FR-16
TC-22	ECG preprocessing (unit)	Synthetic 10 s, 500 Hz, shuffled lead order	Call ECGPreprocessor.transform()	Output shape 12 × 2500; leads in standard order; zero-mean per lead	FR-01
TC-23	Echo inference on valid video	Valid A4C AVI	Run analysis	Mask per frame; EDV > ESV > 0; EF within 0–100; overlay stored	UC-05

ID	Test case	Pre-condition	Steps / input	Expected result	Ref.
TC-24	EF calculation (unit)	—	ejection_fraction(120, 60) and (0, 0)	50.0 and None respectively	FR-07
TC-25	Segmentation failure	Blank / noise video	Run analysis	EF marked unavailable; no EF triage category generated	UC-05 3a
TC-26	Unreadable video	Truncated MP4	Run analysis	echo_results status failed; message “input cannot be processed”	UC-05 2a

Table 29: Test Cases - Fusion and Triage

ID	Test case	Pre-condition	Steps / input	Expected result	Ref.
TC-27	EF band boundaries	Low uncertainty	decide() with EF 50.1, 50.0, 40.1, 40.0, 39.9	Normal, Needs Review, Needs Review, Urgent Referral, Urgent Referral	FR-08, FR-24, OI-3
TC-28	Uncertainty override	EF 62%, EF std > τ_EF	decide()	Needs Review; uncertainty_flag = true	FR-16, SRS 5.2
TC-29	Fusion for verified pair	pair_eligible = true; both results OK	Run analysis	fusion_results row; gate weights sum to 1; triage source = fusion	UC-06, FR-19, FR-40
TC-30	No fusion for single modality	Echo-only case	Run analysis	No fusion row; triage source = echo	UC-06 2a
TC-31	ECG-only case without policy	ECG-only; no policy configured	Run analysis	No triage row; UI shows “No category” with ECG results	UC-06 note, OI-2

Table 30: Test Cases - Results, Case Log and Security

ID	Test case	Pre-condition	Steps / input	Expected result	Ref.
TC-32	View completed results	Owned completed case	Open case	Category, probabilities, EF, overlays, flags and disclaimer displayed	UC-07, FR-25
TC-33	View case still processing	Status processing	Open case	Status shown and polled; no partial result presented as final	UC-07 2a
TC-34	Explanation artifact missing	Artifact removed	Open case	“Explanation unavailable” shown	UC-07 3a
TC-35	Case log filter and pagination	31 owned cases	Filter Needs Review; go to page 2	Only own matching cases; correct counts and pages	UC-08, FR-29
TC-36	Empty case log	New user	Open Case Log	Empty-state message	UC-08 1a
TC-37	Cross-user case access	User B token; case of user A	GET /cases/{id}; direct select via supabase-js	404 from API; zero rows from database	FR-29, FR-33, FR-37
TC-38	Missing / expired / tampered JWT	—	Call /api/v1/cases with each token	401 Unauthorized in all cases	FR-33

ID	Test case	Pre-condition	Steps / input	Expected result	Ref.
TC-39	Secrets not in frontend	Production build	Search bundle for service-role key and DB URL	Not found; only anon key present	FR-38
TC-40	HTTPS enforced	Deployed system	Request http:// URLs	Redirect to HTTPS; HSTS header present	FR-39
TC-41	Signed URL expiry	Artifact URL issued	Reuse URL after 60 s	Access refused	SRS 5.3
TC-42	Audit trail completeness	Completed paired case	Query audit_events for case	Events for creation, file registration, ECG, Echo, fusion and triage with timestamps	SRS 5.2

Table 31: Test Cases - Model Evaluation, Performance and Usability

ID	Test case	Pre-condition	Steps / input	Expected result	Ref.
TC-43	ONNX export parity	PyTorch and ONNX models	Run 200 held-out samples through both	FP32: max probability difference ≤ 0.001; INT8: accuracy change reported	FR-21
TC-44	Patient-level split integrity	Prepared splits	Intersect patient IDs of train/val/test	Empty intersections	FR-02
TC-45	Latency and memory benchmark	Standard and constrained hardware	100 timed inferences per model	p50/p95 latency and peak memory reported (targets TBC)	FR-22, SRS 5.1
TC-46	Calibration report	Validation and shifted test sets	Compute ECE, Brier score and reliability diagrams before/after adaptation	Report produced for ECG class probabilities and Echo reduced-EF probability	FR-15
TC-47	Usability pilot	Volunteer PCPs; consent obtained	Complete scripted tasks: sign in, submit case, interpret result	Task times, SUS score and trust feedback recorded	FR-28

Table 32: Test Cases: Referral Summary Generation

ID	Test case	Pre-condition	Steps / input	Expected result	Ref.
TC-48	Successful referral summary generation	Authenticated and authorized Clinician; selected case has successfully completed ECG and Echo analyses.	Open the case result screen and click Generate Referral Summary.	The successful generation event is recorded in the audit log with the case reference, initiating user, generation timestamp, and operation status.	UC-09, FR-41, FR-44, FR-46, FR-47

ID	Test case	Pre-condition	Steps / input	Expected result	Ref.
TC-49	Single-modality summary generation	Authenticated and authorized Clinician; selected case has one successfully completed modality (ECG-only or Echo-only).	Open the case result screen and click Generate Referral Summary.	The summary includes only the successfully completed modality's available results. The unavailable modality is clearly identified; no findings or values are fabricated.	UC-09, FR-43, FR-45
TC-50	Original analysis data integrity	A case has completed analysis results and is accessible to the authenticated Clinician.	Record the original ECG, Echo, fusion, confidence, uncertainty, and triage outputs. Generate a Referral Summary and compare the stored outputs afterward.	The original analysis outputs, confidence and uncertainty values, and triage decision remain unchanged after summary generation.	UC-09, FR-44, FR-46
TC-51	Unauthorized case access	An authenticated Clinician attempts to access a case for which they do not have permission.	Send a summary-generation request using the inaccessible case reference.	The backend rejects the request. No summary is generated or stored, and the denied access attempt is recorded in the audit log.	UC-09, FR-42, FR-49

ID	Test case	Pre-condition	Steps / input	Expected result	Ref.
TC-52	Missing or invalid authentication token	User is signed out, or the request contains a missing, invalid, or expired Supabase JWT.	Send a `POST /api/v1/cases/{id}/referral-summary` request without a valid authentication token.	The backend rejects the request. No case data is retrieved and no Referral Summary is generated or stored.	UC-09, FR-42, FR-49.
TC-53	Summary request without completed analysis	Authenticated and authorized Clinician; selected case has no successfully completed ECG or Echo analysis.	Open the selected case and attempt to generate a Referral Summary.	The system prevents summary generation and displays an appropriate validation message. No empty or unsupported summary is stored.	UC-09, FR-41, FR-43, FR-49.
TC-54	Stored summary retrieval and access control	A Referral Summary has already been generated and stored for a case.	Retrieve the saved summary using the authorized clinician account, then attempt retrieval using an account without permission to access the case.	The authorized Clinician can retrieve and view the correct saved summary. The unauthorized account is denied access to the summary record.	UC-09, FR-42, FR-46, FR-47, FR-48.
TC-55	Prevention of automatic summary generation	A case has completed at least one relevant analysis; no summary has been requested by a Clinician.	Complete the analysis and inspect the case records without clicking Generate Referral Summary.	No Referral Summary is generated automatically. A summary record is created only after an authenticated and authorized Clinician explicitly requests generation.	UC-09, FR-41, FR-50